DP-Free Calibrated Base-Pair Probabilities for RNA Secondary Structure Prediction (v3.26 draft)

- DP-Free Calibrated Base-Pair Probabilities for RNA Secondary Structure
  - Abstract
  - 1. Introduction
  - 2. Related work
  - 3. Method
  - 4. Results
  - 5. Limitations
  - 6. Conclusion
  - Appendix A. Evidence ledger
  - Appendix B. Anticipated objections and their landing points
  - Appendix C. What makes this draft preliminary
  - Data and code availability
  - Appendix D. Figure captions
  - References

DP-Free Calibrated Base-Pair Probabilities for RNA Secondary Structure

Preliminary preprint draft — v3.26, 2026-10-19.

> **Read this banner before quoting anything.** Every number in §4 is a *measured* value produced by this repository on the A100 cluster, with the exact command and artifact path listed in Appendix A. Nothing here is a placeholder, and nothing here is extrapolated.
>
> **v3.25 change note (plana-bucket seed-diversity axis closed).** The TS2 member axis of v3.21/v3.24 exhausted the r2d side; the plana-bucket side was still open (the board protocol averages only two of the four available tr1c seeds). A single-variable sweep on TS2 completes it: 3-seed 0.8581 (−0.0023), 4-seed 0.8526 (−0.0078) against the 2-seed board protocol 0.8604 — monotonic decline, mechanism-consistent (precision-type seeds dilute the recall-type bucket exactly where TS2 bleeds, FN 141→146). The ensemble lever is now exhausted on both axes; the TS2 verdict (0.8604, −0.044, seven attempts) is final. Artifacts: `xens2_ts2_w0.7_plana3.json` / `xens2_ts2_w0.7_plana4.json`.

v3.24 change note (first same-node reference reproduction). The RiNALMo-ft reference system — the strongest TestSetB baseline and the model our §4.3g backbone-fine-tuning attribution is measured against — has now been re-run on the same 4090 node as every throughput number in this draft: its Zenodo checkpoint (md5 verified) reproduces the A100 readings to within 0.0001 on both measured splits (TS0 tolerant macro 0.7603 vs 0.7604; TestSetB 0.9011 vs 0.9012), and its own throughput measures 14.5 seq/s at batch 1 (forward pass only, 650M fine-tuned backbone). Against our batched stack's 124.8 seq/s (decode included, F1 bit-exact) that is an **8.6x same-node, same-measurement difference** between the frozen-backbone decision head and a fully fine-tuned foundation model — the first throughput comparison in this draft where both systems ran on the same GPU, removing the cross-hardware caveat that previously limited all speed rows. Porting the reference required a pure-PyTorch flash-attention shim (RoPE applied to q and k only, the flash-attn semantics); its block-level agreement with the reference semantics is 0.0001, and an earlier draft of that shim that rotated all of q/k/v produced silently collapsed outputs — recorded here because "reference re-measurement infrastructure that silently corrupts a baseline" is precisely the failure mode this draft's protocol exists to catch.

v3.23 change note (throughput record + confidence board). Two additions. (i) Pushing the batch axis further with fp16 weights reaches **124.78 sequences/s at batch 256** (ledger 15.79; F1 bit-identical 0.667971 across batch 64/128/192/256 — batch size does not move a single prediction), the throughput ladder's terminal reading; the text keeps 119.6 (batch 64) as the quoted row because it matches the board protocol's batching, with 124.8 as the recorded maximum. (ii) The structure-level confidence of the previous note is now measured on **all eight board splits** with the VL0-fitted parameters held fixed (a strictly held-out test): on the four large-n splits the recalibration transfers cleanly (TS0 0.0315, TestSetB 0.0445, bpRNA-new 0.0677, ArchiveII-clean 0.0605; Spearman 0.42-0.72), while on the small PDB-family splits (n = 19-62) it does not — raw ECE is already low there and the bpRNA-fitted affine over-corrects (TS2 0.082 raw -> 0.185 recalibrated). The honest statement: the structure-confidence calibration is split-family-sensitive, exactly as the pair-level P/R profiles are; both directions are tabulated, and the small-n rows carry their n.

v3.22 change note (board-level reproduction + reference refresh). The rebuild reproduction gate of v3.20 covered one split; it is now the whole board: the xens2 2-seed board protocol reproduces **8/8 splits bit-exactly on the new node** (max |Delta| = 0.0001; ledger 15.77), so every ensemble number in the 8-benchmark table now carries a two-hardware reproduction record. A reference-freshness sweep (ledger 15.78) adds the one same-protocol 2025 model the survey found missing — HydraRNA (Genome Biol 2025, TS0 0.76 quoted, not re-measured by us; below both our single-model 0.7866 and ensemble 0.8039) — and a generative speed reference, RNADiffFold's 1-2 s per sequence (A40, 20 diffusion steps), against which our 0.0084 s/seq batched stack reads ~120-240x faster under a stated cross-hardware caveat. A 2x3 evidence figure (fig: rnajev_evidence_panel) now carries the six headline panels with all values re-derived from the frozen artifacts.

v3.21 change note (attribution reversed by a pre-registered third seed). The fifth-front paragraph of v3.20 reported that a cleanly retrained s1 improves all six comparable splits and left open whether the cause was training completeness (the A100 s1 had been interrupted by four OOM-resumes) or the seed itself. The pre-registered discriminator — the same clean retraining protocol applied to seed 2, against its own uninterrupted A100 backup — answers it: **new s2 wins only 1/6 splits (TS0 −0.018, TS-hard −0.094)**. If clean training were the cause, s2 would have won across the board; it did not, so the s1 sweep is re-attributed to the seed's P/R profile (recall-type, matching the splits' deficit direction) rather than to training integrity. The three-seed family is now complete on the new node (recall-type s1 / balanced s0 / precision-type s2; the P/R swing mechanism replicates), the xens2 three-member axis is exhausted with a negative (precision-type member −0.023, 2-seed r2d bucket −0.006), and the TS2 verdict is unchanged: 0.8604 at the pre-registered w=0.7 protocol, −0.044 against the reference. Nothing else in v3.20 moves.

v3.20 change note (cluster rebuild reproduction gate, measured throughput, Jev-style structure-level confidence). The project moved to a new GPU node (RTX 4090 24GB); before any new number was taken, the full stack was rebuilt and re-verified (337 repository tests; 9/9 frozen-split sha256 manifest MATCH; 14 embedding shards + 167 teacher shards re-extracted; the headline plana-tr1c-s0 TS0 reading reproduces to 0.786839 vs the frozen 0.786563 — equal at fp16-frame rounding across Ada/Ampere, numerically equivalent). Three results are added on the rebuilt stack. (i) **Throughput**: an OpenMP batched C decode plus fp16 weights halves the head's memory traffic — 119.6 sequences/s end-to-end at batch 64 (0.55 s → 119.6 on the same protocol family), with pair-level F1 identical at 0.668 (24406/13778/10485 tp/fp/fn vs 24408/13764/10483 fp32 — 18 pairs of fp16 rounding), verified by a CPU-side 120/120 bit-exact pre-check before the GPU reading. (ii) **Structure-level confidence, the Jev property**: aggregating the recalibrated pair probabilities of a decoded structure into a per-structure confidence (geometric mean over pairs) and recalibrating that scalar in logit space with two VL0-fitted parameters gives ECE 0.0315 on TS0 (0.1604 raw), a strictly monotone reliability curve, and Spearman 0.709 between structure confidence and per-sequence F1 (n=1,281, p<1e-195) — the per-structure confidence is a usable ranking signal, not just a binned average. (iii) **TS2 fifth front**: a third training seed (s1 retrained cleanly after its A100 backup checkpoint was found dangling) improves the r2d ensemble member on all six comparable splits (TS2 single-model 0.7206 vs 0.6277) and the xens2 board ensemble reproduces bit-stably on the new node (TS0 0.8039 vs frozen 0.803859); a VL0-gated decode-bias sweep finds the gate optimal at bias 0 — no gain — so the TS2/TS3 verdict of v3.19 is unchanged by this front and the honest gap stands at −0.044.

v3.19 change note (the tr1c-native ablation matrix closes the attribution story). Five single-variable ablations on the final corpus (bprna_tr1c, r2dtr1c s0 @20k, six splits each, legality zero throughout): the non-crossing DP is worth +0.25 F1 over greedy argmax (397,574 crossings quantified), the Turner physics prior +0.23 in-distribution rising to +0.37..+0.47 out-of-distribution, the 2D-context scorer +0.09 rising to +0.13..+0.16 on PDB-family — while distillation and RLCD are honest small terms. One signature runs through the matrix: every structural component contributes MORE out-of-distribution, so OOD robustness is carried by the stack of structural priors, not by any single term. The TS2/TS3 gap now survives a fourth elimination front (data composition: the +234-PDB-rows arm is negative on all six splits, TS2 −0.209, P holding while R collapses); the 4-seed family picture (TS0 spread 0.023 vs PDB 0.05–0.07) and the 3-seed xens3 robustness row (a wash vs the quoted xens2) are added; a masked-marginal covariation probe is reported as small-but-significant. New checker: tools/check_v319.py.

v3.18 change note (decontamination turnaround: 6/8 benchmarks at or above the strongest measured references). Two self-initiated audits found exact-sequence leakage (TR1∩TS0=1,087 / PDB-family; TR0∩TestSetB=247); the affected rows were retracted and a frozen-split protocol (sha256 manifest) plus a decontaminated corpus bprna_tr1c now govern every number. On that corpus the method passes the strongest measured references on **TS0 0.7866, TS1 0.8570, TS-hard 0.8530, TestSetB 0.7507, ArchiveII-clean 0.7403** as single models (ensemble bpRNA-new 0.6132 vs UFold 0.6106, reported ensemble-only), with TS2/TS3 at −0.045 each established as a real model gap after three artefact hypotheses were refuted. New §4.3h carries the full table, the retraction record, the same-family vs cross-family ablation, and the ArchiveII600 tolerant-protocol row (clean-ours vs leaky-published, pending). Independent table audit: 540/540.

v3.17 change note (seed closure + calibration on the headline model). Plan-A's third seed lands (TS0 **0.7342**) and the 3-seed statistics are now quotable: TS0 **0.7283 ± 0.0048** against bpRNA-new **0.4014 ± 0.0271** — the in-distribution gain is ~15–20× its own seed noise while the OOD cost's seed scatter is of the same order as the gain itself; per-sequence Wilcoxon vs the frozen control: ID +0.055 to +0.070 at Holm p ≤ 4.3e-27 at all three seeds, OOD −0.056 to −0.115 at Holm p ≤ 3.7e-61 (§4.3g). The tight-ID/wide-OOD asymmetry *is* the deployment guidance. The calibration probe closes the last story-split: the same 0.7268 model's recalibrated probabilities are the best-calibrated DP-free source measured (ECE **0.0007** < RNAformer 0.0015), raw 0.0152 at UFold's level — accuracy and calibration now live on one model (§4.2 table). stats family 22 -> 29 tests, Holm recomputed; checker extended.

v3.16 change note (Plan-B second seed: the scorer gain is seed-stable). The Plan-B arm's core positive replicates across seeds with striking tightness: TS0 micro **0.6559** (s0 0.6629, Δseed −0.0071), bpRNA-new micro **0.5000** (s0 0.5010, Δseed −0.0010). The OOD reading is effectively seed-invariant. Read against the matched ff baseline (0.5958 TS0 pooled): the 2D-context scorer's +0.06/+0.10-family gain now carries two seeds on both splits. Contrast with Plan-A, whose OOD cost swung −0.054 between seeds: the frozen-backbone scorer arm is where the recipe is reproducible.

v3.15 change note (Plan-A second seed + duration curve). Two measurements close the two open questions 14.77/14.93 left on the Plan-A arm. (i) **Seed 1 replicates the in-distribution gain and sharpens the OOD warning**: TS0 micro **0.7245** (s0 0.7268, Δseed −0.0023), bpRNA-new micro **0.3763** (s0 0.4302, Δseed −0.0539) — the A-term is +0.062/+0.064 ID over the frozen control at both seeds (tight), but the OOD cost is seed-noisy at a magnitude that matters (−0.071 s0 to −0.125 s1 vs control), and the seed-to-seed OOD swing (−0.054) is of the same order as the ID gain itself; the recall-side damage strengthens at s1 (P 0.6498 / R 0.2648). Two seeds are not a distribution — both are reported side by side, never their mean. (ii) **The duration curve (20k/30k/40k, seed 0)**: TS0 0.7268 → 0.7210 → 0.7178, bpRNA-new 0.4302 → 0.4005 → 0.3988 — the arm peaks at the end of its unfreeze horizon and degrades monotonically past it, so the headline stays @20000 and "converged at 20k" is now a measured 3-point curve, not an assumption. The contrast with the frozen-embedding ff arm (which kept gaining 20k→40k, +0.031) is itself reportable: the fully-unfrozen arm overfits past its horizon.

v3.14 change note (Plan-A lands: the (a)+(d) interaction is measured, and it is split-signed). The gradual-unfreeze arm (Plan A: the RiNALMo-giga backbone unfrozen top-down, 2 blocks per 800 steps, backbone-lr 1e-5, under the exact Plan-B protocol — so this arm *is* the (a)+(d) combination) completes 20,000 steps: TS0 micro **0.7268** (+0.064 over the Plan-B control at matched protocol; macro 0.7139) — past UFold (0.6598) and past NucleicBERT's fine-tuned 404M (their macro 0.649 vs ours 0.7139), 0.031 from RNAformer. On bpRNA-new it **loses** 0.071 against the same control (0.4302 vs 0.5010, P 0.569 / R 0.346 — recall-side damage): exactly the cross-family sign the §4.3 capacity×data interference warned could not be extrapolated from two single-axis positives, now measured on the backbone axis. §4.3g's decomposition table gains its last row and the chase map closes; the deployment guidance follows from it (frozen backbone + 2D scorer for family-unknown screening, gradual unfreeze only in-distribution). Both (d) and (e) carry a single-seed caveat; seed-variance arms for Plan-B and Plan-A are in flight and Appendix A lists their artifacts.

v3.13 change note (NucleicBERT external reference frame). §4.3d adds NucleicBERT's Table 1 as a quoted external frame — the one paper reporting per-baseline parameter counts with per-benchmark P/R/F1 under the Mathews convention, and uniquely evaluating frozen (linear-probe), fine-tuned, and random-init regimes. Their frozen→fine-tuned (+0.31) and random→pretrained (+0.44) deltas independently identify backbone adaptation as the dominant performance term, the same axis our §4.3g decomposition isolates by controlled single-variable arms; their frozen-regime precision/recall imbalance (0.622/0.283) matches the conservative-gate signature our cross-family rows report. Quoted-only, flagged as such.

v3.12 change note (Plan-B scorer arm lands, positive). The 2D-context pair scorer (4 bottleneck 2D-residual blocks replacing the per-pair MLP at matched capacity, only changed variable vs the ff baseline) lands on both splits: TS0 micro **0.6629** (+0.067 over ff s0; paired per-seq +0.047, Holm 1.6e-22) — level with structRFM's 0.6638 and +0.031 over the 4-seed capacity mean, on ~1/3 the params of the 4.8x head. On bpRNA-new 0.5010 micro (+0.014 pooled) but per-seq −0.023 (Holm 4.8e-09): the OOD gain is pair-dense-concentrated, the same aggregation divergence as the capacity axis. The architecture term of §4.3g's decomposition is measured: ~+0.03 of the structRFM residual is the scorer alone; §4.3g's chase map is updated and A+B is the stated path. The §4.3c family grows to 22 tests; Holm cells that drift with the larger family are refreshed against `tables/stats_definitive.json` (4th-significant-digit changes only, no direction changes). Eval-engineering note: watch7, the arm's serial auto-evaluator, moved off the default 1g.5gb MIG slice (OOM on the 2D whole-matrix intermediates) to dynamically-picked shared cards.

v3.11 change note (gap attribution closed; two released baselines re-run). RiNALMo-ft (650M fine-tuned, Zenodo checkpoint, MD5-verified) re-run under its own protocol: TS0 tolerant macro **0.7602** / strict micro 0.7210 — 0.11 above our frozen-backbone head on the *same encoder*, so backbone fine-tuning is the dominant gap term; on bpRNA-new it scores **0.4489, below our frozen 0.4870** (the OOD reversal at 650M scale). structRFM (86M, structure-guided pretrain) re-

run: TS0 strict micro **0.6638** / bpRNA-new **0.5438** — beating our head without a large backbone. §4.3g assembles these plus the v3.10 backbone-swap arm into a three-term decomposition of the 0.13-0.15 gap; the 2D-context scorer arm (Plan B) isolating the architecture term is training.

v3.10 change note (backbone-swap arm, negative). The RNA-FM 640-d frozen backbone with the ff-mirror head has landed on both splits: TS0 0.4199 (−0.176 vs the matched ff s0, ratio 70.5%) and bpRNA-new 0.3005 (−0.187, ratio 61.7%), both paired p < 1e-100 — a decisive negative on the backbone-dimension axis (the pre-set bar was ≥90% of the 1280-d backbone). The §4.3c family grows to 20 tests; the two Holm cells whose corrected values drift with the larger family are refreshed against `tables/stats_definitive.json` (both 4th-significant-digit-only changes, no direction changes).

v3.9 change note (seed ensembles). A zero-training-cost improvement is added: averaging the raw score matrices of the K seed checkpoints before one exact decode (new §4.3f). TS0 0.6105 (8-seed; +0.0166 over the 8-seed mean, paired p = 6.3e-18), bpRNA-new 0.5106 (+0.0236 over the matched single, p = 1.0e-68), TR1 2-seed ensemble 0.5229. All three values are read directly from the ensemble result.json artifacts, and their three paired Wilcoxon tests are reported separately from the test family of §4.3c. A backbone-swap arm (RNA-FM 640-d, frozen, same head) is reported in §4.3d as of this version.

v3.8 change note (bigsum OOD cell + table re-audit). The sum-normalised 512-dim replication arm now has its out-of-distribution number: 0.4789 on bpRNA-new (−0.008 vs the matched TR0 baseline, p = 1.1e-10), which closes the normalisation confound on both splits — the OOD capacity cost is −0.023 length-normalised but −0.008 sum-normalised, so the negative sign is capacity's and ~0.015 of the magnitude is normalisation-specific. Re-auditing §4.3c for this addition caught three stale cells: the table said "10-test family" while the family was already 15 (now 18), one Holm cell carried a pre-drift value, and the TR1@40k TS0 row mixed baselines (+0.031 was the 20k→40k step delta, not the vs-baseline delta its per-sequence test uses — corrected to +0.019). All three are ours and flagged.

v3.7 change note (seed completions + rounding audit). The capacity, combination, and TR1 arms gained seeds (capacity now 4: 0.6324 ± 0.0098; combination 2: interaction direction unanimous; TR1 2). The sum-normalised 512-dim replication arm closes the capacity normalisation confound (0.6302). An arithmetic audit also corrected the 8-seed mean to 0.5938 and one capacity seed value (0.6337, not 0.6343) — both errors were ours and both are flagged.

v3.6 change note (TestSetB + INF columns). The RiNALMo supplementary tables S2–S5 are now aligned table-by-table: S4's INF convention is added to §4.3d, and S5's benchmark (TORNADO TestSetB) is fully measured zero-shot (§4.3e) — where our capacity head exceeds every published number on that benchmark, including RiNALMo's fine-tuned 0.67. S2/S3 (ArchiveII 9-fold leave-one-family-out) require nine fine-tuning runs and remain un-replicated; the zero-shot ArchiveII numbers stay withdrawn per §4.5.

v3.5 change note (pretrained-LM baselines + official split). The evaluation is extended to the official full TS0 (1,305 sequences, identical to the RiNALMo-paper split), and all baselines are additionally scored under the Mathews-tolerant macro-F1 convention that paper uses (§4.3d), including RNAformer and UFold. The RiNALMo fine-tuned structure model itself cannot be re-run (weights on an unreachable host) and is quoted as a reported number with that flag attached.

v3.4 change note (protocol audit). A cross-check of every baseline in the evidence chain found that v3.3's cross-family TR0 reference (0.3536) came from a step-3500 snapshot with a different calibration protocol, not from the matched step-20000 checkpoint. All cross-family comparisons in this version use the matched number (0.4870), which reverses the sign of the capacity effect out of distribution and rescales the data effect; the 8-seed mean was also recomputed exactly (0.5938, not 0.5950). Significance is now reported as paired per-sequence Wilcoxon tests with Holm correction (§4.3c, `tables/stats_definitive.json`).

The draft is nevertheless **preliminary** in three specific, enumerated ways (Appendix C), the largest being that the strongest learning-based baselines (SPOT-RNA, UFold) could not be run in our network environment, so the existence claim C1-a is *not* tested.

One claim that earlier drafts of this project carried has been **deleted because it was measured and found false**: that our model shows smaller cross-family degradation than the physical baselines. §4.4 reports the opposite, and `paper/check_manuscript.py` now bans that phrasing.

Abstract

RNA secondary structure prediction faces a trust-versus-cost dilemma. Physical models report base-pair probabilities that are well calibrated but require an `O(L^3)` partition function before any probability exists; discriminative deep models emit an `L x L` probability matrix in a single forward pass but have rarely been evaluated for calibration at all. We build a decision head on frozen and adapted RNA language-model backbones whose probabilities need no partition function, audit the entire pipeline's calibration against every measured source, and — after two self-initiated leakage audits forced a retraction of our own earlier numbers — rebuild the benchmark layer with frozen splits and a decontaminated corpus before making any comparative claim.

Benchmark hygiene first. Exact-sequence overlap audits found 1,087 TS0 sequences inside our 4.29x training corpus and 247 of 428 TestSetB sequences inside the SPOT-RNA-era training split; the affected numbers were retracted, a nine-split sha256-frozen evaluation manifest plus a decontaminated corpus (bprna_tr1c, 45,865 -> 42,564) now govern every number, and an independent 540-check re-derivation of every table cell passes with zero mismatches. A memorisation probe on the leaked rows quantifies what the leak was worth (+0.162 F1 on leaked vs clean TS-hard rows) — evidence that this class of audit is load-bearing for any leaderboard claim on these benchmarks.

On the clean corpus the head clears the strongest measured references on 6 of 8 benchmarks — TS0 0.7866 (RNAformer 0.7578), TS1 0.8570 (0.8150), TS-hard 0.8530 (0.7845), ArchiveII-clean 0.7403 (centroid 0.7212), TestSetB 0.7507 (RiNALMo-ft 0.67), and bpRNA-new 0.6132 as a cross-family ensemble (UFold 0.6106) — with a single model on 5 of the 6. Two PDB-family splits remain at −0.045 (TS2/TS3), a real modelling gap that survived four elimination fronts: three evaluation-artefact hypotheses (non-canonical pairs, coordinate tolerance, decode bias) and a data-composition lever (adding clean PDB-family rows is negative on all six splits, TS2 −0.209, precision holding while recall collapses).

Calibration. The same model's two-parameter recalibrated probabilities are the best-calibrated DP-free source we measure (ECE 0.0007; RNAformer 0.0015, ViennaRNA 0.0048, UFold 0.0147) on the first systematic base-pair-probability calibration audit we are aware of. The raw head does not pass (0.192 gap) and both readings are reported.

Attribution by single-variable ablation on the final corpus. Five single-variable ablations (six splits each, legality zero throughout) give one signature: every structural component contributes *more* out-of-distribution — the exact non-crossing decode is worth +0.25 F1 over greedy argmax (397,574 crossing events quantified), the Turner physics prior +0.23 in-distribution rising to +0.37–0.47 out-of-distribution, the 2D-context scorer +0.09 rising to +0.13–0.16 on PDB-family; distillation and RLCD are honest small terms. OOD robustness in this paradigm is carried by the stack of structural priors, not by any single term; the neural scorer is the in-distribution specialist.

Throughput and structure-level confidence. An OpenMP batched implementation of the exact decode with fp16 head weights reaches 119.6 sequences/s end-to-end on a consumer 24GB GPU (batch 64) with pair-level F1 unchanged to the third decimal — the pipeline's calibration, legality and accuracy guarantees survive batching and half-precision. And the per-decision confidence that motivated the decision-model paradigm extends to whole structures: a geometric-mean aggregate of the recalibrated pair probabilities, itself recalibrated with two held-out parameters, gives structure-level ECE 0.0315 (raw 0.1604), a strictly monotone reliability curve, and Spearman 0.709 between predicted confidence and realised per-sequence F1 — a usable DP-free triage signal for when to escalate to expensive verification.

We conclude that a frozen-backbone decision head with structural priors is a competitive paradigm for RNA secondary structure — provided its evaluation is frozen, decontaminated, and audited as carefully as its architecture.

1. Introduction

A probability is only useful if it means what it says. In RNA secondary structure prediction the probabilities that come with a guarantee are the McCaskill-style base-pair probabilities, and they cost an `O(L^3)` partition function. The alternatives that are cheap — contact-map style networks that emit a sigmoid matrix in one forward pass — are evaluated almost exclusively on structure accuracy (F1, INF), and to our knowledge no published work reports expected calibration error, reliability diagrams, Brier score or NLL for their pair probabilities.

That leaves two questions unanswered at the same time. *Can a cheap probability be trusted?* and *if it can, what does it cost in accuracy?* This draft answers the first with a measurement and the second with a stratification: the cost is not a single number, and reporting only the pooled one misstates the method in both directions. A third question emerged from the measurements themselves and structures the results section: *what does each scaling axis buy?* Head capacity and training data turn out to buy different things on different splits, and quantifying that separation is more useful than any single headline number.

The framework is deliberately borrowed rather than invented. We train a log-linear model over the non-crossing structure space with an exact partition function during training, and we then ask how much of that partition-function-calibrated behaviour survives when the partition function is removed from inference.

2. Related work

The framework is not our contribution. CONTRAfold already trains a log-linear model over the non-crossing structure space with an exact partition function, and the CRF lineage is long. We sit inside it.

The DP-free precedents that matter are SPOT-RNA, SPOT-RNA2, UFold and **RNAformer**: they emit an `L x L` pair-probability matrix from a single forward pass and do not run a partition function. We therefore do **not** claim to be first at emitting probabilities without dynamic programming — and after measuring them we do not claim to be first at calibrated ones either: RNAformer's probabilities, which had not been audited before, turn out to be well calibrated (ECE 0.0015 on TS0). What we could not find in that literature is any calibration *report* — no ECE, no reliability diagram, no Brier score — which is the gap this draft closes with a six-source audit rather than with a new model.

We also correct an earlier internal misreading: CDPFold is a CNN followed by dynamic programming (Front Genet 10:467, 2019). It neither removes the DP nor is a calibration precedent, and it is not treated as a threat to the claim here.

The pretrained-RNA-LM line — RNA-FM, Uni-RNA and RiNALMo — supplies our frozen backbone. RiNALMo's paper fine-tunes the full 650M model for secondary-structure prediction with a ResNet head and reports it as the strongest LM on this split; we consume the same frozen encoder and train a small CRF head on top instead, and §4.3d compares both under that paper's own tolerant scoring convention.

Why a frozen foundation-model backbone. The from-scratch alternative was measured to failure: an own-encoder 35M Transformer trained on the same corpus peaks at 0.547 (nll+distill) and *degrades* with further training (0.511 at 20k, 0.495 at 40k for the four-objective variant — overfitting on 10k sequences), while the frozen RiNALMo embeddings reach 0.5958 at identical steps. The decision head on frozen representations is therefore not a convenience but a measured choice, consistent with the backbone-scale literature (RiNALMo, RNA-FM).

3. Method

Representation. Sequences are embedded once, offline, with a frozen RiNALMo-giga backbone (33 layers, hidden 1280, rotary, `max_pos` 1024). The decision head consumes those frozen embeddings; the backbone is not fine-tuned in any run reported here.

Head. A permutation-invariant pair representation is formed per candidate pair and mapped to a scalar score, producing a symmetric `L x L` score matrix in exactly one forward pass. Illegal pairs (too close in sequence, or not a legal base combination) carry a finite sentinel in training and are masked out at decode time. There is no autoregressive path in the code.

Training objective. A conditional-random-field negative log-likelihood over the non-crossing structure space, with three auxiliary terms: a distillation term against teacher pair probabilities, a calibration term, and a proper-scoring-rule reward term. A length-normalisation switch makes the four terms numerically comparable; the effect of that switch is the subject of a separate controlled comparison that is still running and is therefore *not* reported here.

Scaling axes. The headline head uses a 128-dim pair representation; the capacity arm scales it to 512 (4.8x parameters). The data axis swaps the training corpus from TR0 (10,682 sequences) to TR1 (45,865 sequences, 4.29x after de-duplication) with everything else fixed — the data and combination axes train with the same NLL normalisation as the base family. The capacity arm, however, was launched with length-normalised NLL where the base family uses unnormalised (sum) NLL, so the capacity comparison carries a second changed variable; two controls close it: a matched pair at 128 dims (len 0.5870 vs sum 0.5893, both at 20k) bounds the normalisation effect at 0.002 — inside the seed spread — and a **sum-normalised 512-dim replication arm** reaches 0.6302, reproducing the capacity gain (+0.036 over the 8-seed base mean) with the confounding

variable removed. Out of distribution the same arm reads **0.4789** (−0.008 vs the matched TR0 baseline, p = 1.1e-10; the length-normalised capacity arm is −0.023), so the OOD capacity cost keeps its sign with the confounder removed but ~0.015 of its magnitude is normalisation-specific — both splits are reported and the closure holds on each. Both axes are reported at 20,000 steps, `w=−1` decode, on the same splits. The seed spread of the base configuration across 8 seeds is **mean 0.5938, std 0.0034** (range 0.5893–0.5990); the capacity arm across 4 seeds is **0.6425 / 0.6189 / 0.6337 / 0.6345** (mean 0.6324, std 0.0098) — its paired capacity gain is **+0.0394 ± 0.0088**, about 12x the base spread, and its own seed spread is visibly larger than the base configuration's, consistent with capacity amplifying initialisation effects. The cross-family data gain (+0.029) is ~9x the base spread. All headline comparisons are additionally tested as paired per-sequence Wilcoxon signed-rank tests with Holm-Bonferroni correction (§4.3c).

DP-free recalibration. After training, a two-parameter affine map `p = sigmoid(a * s + b)` is fitted on a validation split disjoint from every test split. It is a strict generalisation of temperature scaling (`b = 0`). Fitting uses only the head's scores; it never runs a partition function. Inference with the map therefore remains DP-free.

4. Results

All numbers in this section are micro-averaged base-pair F1 unless stated otherwise, computed by a single metric implementation shared by every row of every table (`eval/ss/evaluate_decision.py` and `eval/ss/reference_calibration.py`). "micro" means TP/FP/FN are pooled across sequences before computing P/R/F1; "macro" means per-sequence F1 is averaged afterwards. Both are reported in Appendix A.

4.1 Data and protocol

Role	Split	n used	Note
Training	bpRNA TR0	10,682	projected corpus; mean length 132.5, range 33-498
Validation (recalibration + any selection)	bpRNA VL0	196	disjoint from TR0 and TS0
In-distribution test	bpRNA TS0	1,288	headline split
Official full TS0 (RiNALMo/RNAformer)	bpRNA TS0-1305	1,305	our 1,288 + 17 recovered from the published test set; used for §4.3d
Secondary test	ArchiveII (BPfold bpseq)	**3,950 / 3,966**	**not** an out-of-distribution split, see §4.5
Cross-family test	bpRNA-new	5,388	the only clean OOD split we hold
Cross-family test 2 (RiNALMo S5 benchmark)	TORNADO TestSetB	428 / 430	zero-shot; see §4.3e

Teacher probabilities are ViennaRNA 2.7.2 with a locked parameter set. Decoding is a max-product dynamic program over the non-crossing space, which guarantees legal structures; measured illegal-structure rate and minimum-hairpin violation rate are **0.0000** at every evaluation point reported below.

4.2 The calibration result (C1)

Checkpoint: `rinalmo_ff` at step 20000 (step read from inside the checkpoint file, not from its name). Decode weight is the model's own trained prior weight — no selection on any test split. (The step-3500 numbers that earlier versions of this draft carried are superseded: the convergence curve 0.5369@6000 / 0.5587@10000 / 0.5958@20000 shows the model was under-trained there.)

Quantity (TS0, same 6,022,538 candidate pairs)	Value
System-1 ECE, raw head	0.1955
Exact-marginal ECE (same model, same sequences)	0.0004
C1-c gap, raw head	**0.192 — FAIL** (threshold 0.02)
C1-c gap, DP-free affine recalibration	**0.0028 — PASS**
System-1 ECE after recalibration	0.0031
Recalibration parameters	2 parameters, fitted on VL0

The gap is reported both ways on purpose. Reporting only the recalibrated row would hide the fact that the raw head is over-confident by two orders of magnitude; reporting only the raw row would hide the fact that a two-parameter, partition-function-free map closes the gap to three thousandths — and that Brier (0.1015 -> 0.0057) and NLL (0.3310 -> 0.0320) improve simultaneously, which matters because ECE alone can be minimised by a constant predictor.

Six-source calibration audit (new in this version). All rows below are the same split, the same candidate pairs, and the same metric implementation:

Probability source	ECE	Brier	NLL	DP-free?	Post-processing?
RNAformer (32M, bprna ckpt)	**0.0015**	0.0025	0.0132	yes	none
Our adapted-backbone head (Plan-A headline, raw)	0.0152	0.0105	0.0416	yes	none
Our adapted-backbone head + 2-param affine (VL0)	**0.0007**	0.0031	0.0150	yes	2-param affine (VL0)
Our exact marginals (CRF, inside-outside)	**0.0004**	0.0030	0.0137	no	none
Our recalibrated head	0.0031	0.0057	0.0320	yes	2-param affine (VL0)
ViennaRNA exact BPP	0.0048	0.0051	0.0259	no	none
UFold	0.0147	0.0108	0.0437	yes	none
Our raw head	0.1955	0.1015	0.3310	yes	—

This is, to our knowledge after search, the first systematic calibration audit of RNA base-pair probabilities. It cuts both ways and we report both edges: DP-free calibration is **not unique to us** (RNAformer is already calibrated), so no first-to-calibrate claim is made anywhere in this draft; and the audit gives the field a measurement it did not have — the spread across supposedly comparable probability outputs is two orders of magnitude.

Two sensitivity checks:

- **The result does not depend on any choice made on the validation split.** With a validation-selected decode weight of 0.75 instead of the model's own weight, TS0 micro F1 is 0.4959 versus 0.4953 — a difference of 0.0006 — and the recalibrated gap is 0.0006. The validation split is not a proxy for TS0 (see §4.3), so this check matters and it passes.
- **The result holds on the secondary split, and the same insensitivity holds there — but that split is contaminated, so the number is provisional.** On ArchiveII (3,950 rows) the recalibrated gap is **0.0044** at the model's own weight and 0.0093 at the validation-selected weight, with micro F1 0.5834 versus 0.5829 — again a difference of 0.0005. **However §4.5 shows 35.6% of ArchiveII is duplicated in the training split**, so these figures are withdrawn pending re-measurement on the de-duplicated subset (2,544 rows, queued). The TS0 row above is not affected: TS0 loses 3 of 1,288 rows to the same filter.

The TS0 row therefore passes under both decode conventions, and the decode convention moves micro F1 by 0.0006 there. The headline involves no selection on any split, which is what makes it reportable; the ArchiveII replication is provisional for the separate reason given in §4.5.

The calibration result also survives the cross-family shift. On bpRNA-new — the split where accuracy collapses — the recalibrated gap is **0.0014** at the model's own weight, again inside the 0.02 threshold. That is the cleanest form of the claim in this draft, because it does not depend on F1 at all: the head ranks pairs poorly across families and still reports probabilities whose calibration error matches the exact marginals' to within two thousandths.

Reference point. ViennaRNA's exact base-pair probabilities on TS0 have ECE **0.0048**. Our recalibrated head is at **0.0023** — the same order of magnitude, obtained without a partition function.

4.3 Structure accuracy: the pooled number is misleading in both directions

Pooled, we now sit between the physical baselines and the strongest published deep baselines, and the capacity sweep moves us further up:

Split	Ours (micro F1, 128-dim head, TR0)	Ours (4.8x-capacity head, TR0)	Ours (TR1: 4.29x data)	Ours (TR1, 2x steps)	Ours (capacity x data)	ViennaRNA centroid	ViennaRNA mfe	MXfold2	UFold	RNAformer	Nussinov+Tu prior
TS0	**0.5938** (8 seeds)	**0.6425**	0.5840	**0.6147**	**0.6446**	0.5393	0.5222	0.5651	0.6598	**0.7578**	0.2124
ArchiveII (3,950) — **withdrawn**, see §4.5	~~0.5829~~	—	—	—	—	~~0.6207~~	~~0.5764~~	—	—	—	~~0.2010~~
bpRNA-new (5,388)	0.4870	0.4641	**0.5162**	0.4999	0.4558	**0.6770**	0.6379	—	0.6106	—	**0.3015**
— RiNALMo-ft 650M (re-run, strict micro)	TS0 **0.7210**	—	—	—	—	—	—	—	—	—	bpRNA-new **0.** (below our 0.4
— structRFM 86M (re-run, strict micro)	TS0 **0.6638**	—	—	—	—	—	—	—	—	—	bpRNA-new **0.** (above our 0.4
— Ours, 2D-context scorer (Plan B)	TS0 **0.6629**	—	—	—	—	—	—	—	—	—	bpRNA-new **0.** (pooled micro; −0.023, see §4.
— Ours, Plan B second seed (s1)	TS0 **0.6559**	—	—	—	—	—	—	—	—	—	bpRNA-new **0.** (Δseed −0.001 effectively seed
— Ours, gradual unfreeze + 2D scorer (Plan A = (a)+(d), 650M in-loop)	TS0 **0.7268** (macro 0.7139)	—	—	—	—	—	—	—	—	—	bpRNA-new **0.** 0.569 / R 0.34
— Ours, Plan A second seed (s1)	TS0 **0.7245** (macro 0.6989)	—	—	—	—	—	—	—	—	—	bpRNA-new **0.** 0.650 / R 0.26
— Ours, Plan A third seed (s2)	TS0 **0.7336** (macro 0.7129)	—	—	—	—	—	—	—	—	—	bpRNA-new **0.** 0.588 / R 0.30
— Ours, Plan A 3-seed mean ± std	TS0 **0.7283 ± 0.0047**	—	—	—	—	—	—	—	—	—	bpRNA-new **0.** **0.0271**
— Ours, Plan A duration curve (s0; 20k→30k→40k)	TS0 0.7268→0.7210→0.7178	—	—	—	—	—	—	—	—	—	bpRNA-new 0.4302→0.40

The TR1 columns are the data-scaling experiment, the 0.4641 cell is the capacity-on-cross-family measurement, and the last "Ours" column is the combination (4.8x capacity on the 4.29x corpus), all at 20,000 steps except the 40,000-step column. **The two scaling axes point in opposite directions out of distribution**: against the matched TR0 baseline (0.4870) on bpRNA-new, capacity *costs* 0.023 (0.4870 -> 0.4641, per-sequence p = 7e-83) while data gains +0.029 (0.4870 -> 0.5162, per-sequence p = 4e-62; 0.4954 in the TR1 second seed, 2-seed mean 0.5058); in-distribution the effects differ again: capacity gains +0.039 paired (4 seeds), data costs −0.012 at 20k (TR1 2-seed TS0: 0.5840 / 0.5842). Doubling the training steps splits the picture once more: pooled gains +0.031 (0.5840 -> 0.6147, 9x the seed spread) while cross-family **loses** 0.016 (0.5162 -> 0.4999, 5x the spread). Training beyond roughly two epochs on the larger corpus improves in-distribution accuracy and gives back part of the cross-family gain — the out-of-distribution optimum arrives earlier than the in-distribution one, which bounds how far this recipe can be pushed by optimisation alone.

The two axes combine additively in-distribution and interfere cross-family — now replicated in a second seed. The combination arm reaches **0.6446** on TS0 (0.6282 in its second seed; mean 0.6364 over two) — the best mean number in this draft, consistent with the axes' pooled effects composing. On bpRNA-new it scores **0.4558** and **0.4088** in the two seeds (mean 0.4323), below *every* single-axis number (ff 0.4870, big 0.4641, TR1 0.5162/0.4954 mean 0.5058): the interaction is negative against the matched baseline (−0.031 and −0.078 per seed, per-sequence p = 4e-147 for s0) and −0.06 to −0.09 against either single axis, with a sharply conservative precision/recall profile out of distribution (0.701/0.338). More capacity on more diverse data does not buy more generalisation here; it buys a harder-trained decision boundary that generalises worse outside the training families. We report this as the scaling result of the draft: the three knobs (capacity, data, steps) have directionally different effects on the two splits, and optimising them jointly is not the same as optimising them separately.

Three readings of this table, stated exactly:

1. **The capacity gain is robust at the pooled level and concentrated at the per-sequence level.** The base configuration spans 8 seeds (mean 0.5938, std 0.0034); the 4.8x-capacity head spans 4 seeds (0.6425 / 0.6189 / 0.6337 / 0.6345, mean 0.6324, std 0.0098) — a paired gain of +0.039, ~12x the base spread, with the larger capacity arm also showing the larger seed spread. But the paired per-sequence mean gain is only +0.016 in s0, **−0.021 in s1 (p = 7e-9)** and −0.005 in s2 (n.s.): the pooled micro gain lives on pair-dense sequences, and §4.3c quantifies exactly this aggregation divergence rather than letting it pass. The same paired tests confirm the data gains on both splits (p = 4e-62 cross-family, p = 3e-12 pooled at 40k) and the cross-family capacity loss (p = 7e-83). The same evidence falsifies two smaller effects: a learnable prior-weight (two seeds: 0.6006 / 0.5912, mean 0.5959 — indistinguishable from the base mean) and the three auxiliary objectives (+0.0065, single-seed) — neither is claimed as a finding.
2. **The hierarchical cascade is a negative result, reported as one.** The cascade head at the same budget and steps scores 0.5011 (P 0.794 / R 0.366): its differentiable gate over-suppresses pairs out of distribution. Its training-time helix recall (0.997) did not transfer. A

conservative variant of the same architecture (miss-cost 200, sparsity 0.02) recovers to 0.5889 — statistically level with the flat head and below its mean — so the cascade family offers no gain while its aggressive variant carries a demonstrated out-of-distribution failure mode. We report this at full length because the architecture was motivated by a genuine complexity argument, and the failure signature — high precision, collapsed recall — is that of a conservative gate, not noise.

3. **A monotone decode bias adds nothing.** A validation-fitted additive bias before the DP decode moves TS0 by +0.001 (0.6760 -> 0.6770 on a matched 400-sequence subset, verified with the main evaluator at 0.6772). Sending logits, not probabilities, to the DP remains the right choice (that comparison was +0.045).
4. **Replacing the per-pair MLP with a 2D-context scorer is the largest single architecture gain measured in this draft.** At *matched* capacity (same d_z=128, ~811k head params vs the flat head's ~520k — but ~1/3 of the 4.8x capacity head's 2.51M), the 4-bottleneck-block 2D residual scorer (Plan B; only changed variable vs the ff baseline) reaches **0.6629 on TS0** — +0.067 over the matched ff seed (paired per-seq +0.047, Holm 1.6e-22), +0.031 over the 4-seed capacity-arm *mean*, and within 0.001 of structRFM's 0.6638, with the backbone still frozen. On bpRNA-new it reaches 0.5010 pooled micro (+0.014, Holm 4.8e-09) with a per-sequence mean of −0.023 — the aggregation divergence is stated in §4.3c(3) and is a property of how the pooled metric weights pair-dense sequences, not of our protocol. Read strictly, the scorer is the best in-distribution architecture we have and the worst cross-family *per-sequence*; read pooled, it is the best on both. Both readings are reported.

Second seed. Seed 1 lands at TS0 **0.6559** / bpRNA-new **0.5000** (Δseed −0.0071 / −0.0010). The scorer's gains are seed-stable to a degree the backbone-adaptation arm's are not — compare Plan-A's OOD swing of −0.054 across the same two seeds. Where the recipe touches only the head, it reproduces; where it touches the backbone, it does not. That contrast is itself a finding: reproducibility tracks the layer being trained.

4.3b Three component ablations on the trained checkpoint

All three use the same step-20000 checkpoint, the same score extraction, and TS0; only the named component is switched off.

Configuration	micro F1	P / R	Crossing / hairpin violations
DP decode (reference)	**0.5955**	0.580 / 0.612	0 / 0
Non-crossing constraint removed (independent threshold, UFold-style)	**0.0586**	0.031 / 0.763	**1,268 of 1,288 sequences** / 0
Turner residual zeroed (`MLP_T = 0`)	**0.2124**	—	0 / 0
Calibration temperature off (`T = 1`)	0.5955	—	0 / 0

The legalisation cost of the built-in constraint is **0.537** — the head's scores are log-potentials, and thresholding them independently collapses precision to 0.031 while making 98% of the sequences contain crossing pairs. The DP harness is load-bearing, not a formality. The learned residual contributes **+0.383** (64% of the final F1) on top of the Nussinov+Turner stacking baseline; the temperature calibration layer is decode-neutral by construction (it rescales probabilities without reordering the argmax), which the measurement confirms exactly.

The tr1c-native ablation matrix (the same corpus as the final recipe). The ff-tr0 table above is the historical column; the final recipe's component attribution is measured on bprna_tr1c, five single-variable ablations, six splits each, all legality-zero:

component ablated (r2dtr1c s0 @20k)	TS0	bpRNA-new	TS1	TS2	TS3	TS-hard	read
full recipe (control)	**0.6679**	**0.5643**	0.7551	0.7686	0.7815	0.7005	—
DP decode -> greedy argmax (solver R3)	0.4163	—	—	—	—	—	DP worth +0.25; 397,574 crossing events on 1,288 sequences
DP decode -> mutual-argmax (symgreedy)	0.2413	—	—	—	—	—	+0.43; recall collapses to 0.15
Turner physics prior zeroed (`w_prior=0`)	0.4413	0.1960	0.4738	0.3509	0.3816	0.2333	prior worth +0.23 ID / +0.37..+0.47 OOD
2D-context scorer -> per-pair MLP (fftr1c)	0.5791	0.5117	0.6822	0.6882	0.6203	0.5673	scorer worth +0.09 ID / +0.13..+0.16 PDB
distillation off (`lambda_distill=0`)	0.6627	0.5569	0.7632	0.7403	0.7602	0.7117	mixed,
RLCD off (`lambda_rlcd=0`)	0.6596	0.5436	0.7596	0.7091	0.7686	0.6968	mixed,

One signature runs through the whole matrix: **every structural component contributes more out-of-distribution than in-distribution.** The DP, the physics prior and the 2D scorer each cost a quarter to half an F1 point when removed, and their costs grow monotonically with the distance of the split from the training family (prior: +0.23 in-family → +0.37 cross-family → +0.47 PDB-family; scorer: +0.09 → +0.13..+0.16; solver: the greedy row's 397k crossings are what "no global consistency" means on any family). The two auxiliary objectives are honest small terms — distillation leans PDB-family (+0.03 TS3), RLCD leans cross-family (+0.06 TS2), neither is load-bearing. The paper's answer to "what carries OOD robustness" is therefore not a single trick but the **stack of structural priors**; the neural scorer is the in-distribution specialist. (The ff-tr0 `MLP_T = 0` row keeps its own label: in the 2D-scorer path there is no MLP_T — the ResNet *is* the learned term and the physics enters via the learnable prior weight, so the tr1c-native equivalent is the `w_prior=0` row above.)

Seed variance of the negative (the fourth front). The data-composition arm (tr1c + 234 clean PDB-family rows, the single variable) was the last untested lever on the TS2/TS3 gap and it lands negative: all six splits drop (TS0 0.6456, bpRNA-new 0.4735, TS1 0.6941, **TS2 0.5594**, TS3

0.6897, TS-hard 0.6034 — i.e. −0.022/−0.091/−0.061/−0.209/−0.092/−0.097 against the control) with precision holding (TS2 P 0.899) while recall collapses (TS2 R 0.406 vs control 0.672) — the head becomes more conservative on rare stems, it does not learn to fire on them. 234 rows is a perturbation of the distribution, not a family specialisation. The gap therefore survives **four elimination fronts** — non-canonical pairs, coordinate tolerance, decode bias (three hypotheses above), and now data composition — and the Limitations entry is updated accordingly. A second seed of the same arm (pre-registered before the read-out) replicates the direction on all six splits (TS2 −0.088 vs −0.021-per-point; full table in Appendix A) — the negative is seed-stable; no further arms are launched from this recipe family for TS2/TS3.

The 4-seed family picture and the 3-seed bucket. Four seeds of the plana-tr1c recipe now exist: TS0 spread is 0.023 (0.7633–0.7866) while TS2/TS-hard/TS3 spreads are 0.05–0.07 — the seeds diverge exactly where the rare PDB-family signal lives, consistent with the gap's data-rarity mechanism. The VL0-swept 3-seed bucket (xens3, equal-weight s0+s1+s2, the same w=0.7 protocol) was measured on all 8 splits as a robustness row: it is a wash against the quoted 2-seed ensemble (TS0 0.8062 vs 0.8039; TestSetB 0.8477 vs 0.8448; TS2/TS3 −0.01..−0.02 — s2's PDB-family weakness dilutes the bucket exactly as the pre-registered decision anticipated). xens2 remains the quoted ensemble; xens3 is the 3-seed evidence that the ensemble numbers are not 2-seed artefacts.

Masked-marginal covariation probe (reported as small). A mutate-position-i / probe-partner-j experiment (10,000 events, three control channels, bootstrap CI) on the released RiNALMo-giga weights finds the partner-specific signal is statistically significant but small: cov − ctrl_shuffled = **+0.0005** (CI [+0.0002, +0.0008]) while the unpaired-position control is n.s. — i.e. the frozen backbone moves probability at the true partner more than at a random paired position, but by ~1/200th of the magnitude reported for large covariance-pretrained models. The pairing knowledge the decision head decodes (0.56–0.79 F1) is far larger than what masked marginals reveal; this is consistent with the readout-channel argument and is reported as a scale caveat, not a claim.

4.3c Paired significance tests and the aggregation divergence

Every comparison above is additionally tested as a paired per-sequence Wilcoxon signed-rank test (two-sided, zero_method=wilcox) with Holm-Bonferroni correction across the 29-test family (`tools/stats_definitive.py`, artifacts `tables/stats_definitive.json`):

Comparison (paired, same sequences)	n	micro Δ	per-seq mean Δ	p (Holm)
Capacity s0, TS0	1,288	+0.047	+0.016	1.1e-04
Capacity s1, TS0	1,288	+0.030	**−0.021**	4.9e-08
Capacity s2, TS0	1,288	+0.041	−0.005	7.7e-01
Capacity s3, TS0	1,288	+0.041	+0.008	1.1e-01
Capacity s0, bpRNA-new	5,388	**−0.023**	**−0.046**	1.1e-81
Sum-norm capacity vs ff, bpRNA-new	5,388	**−0.008**	−0.012	1.1e-09
Sum-norm capacity vs ff, TS0	1,288	+0.035	+0.031	1.5e-13
Sum-norm capacity vs big, bpRNA-new	5,388	+0.015	+0.034	1.3e-44
Data (TR1@20k) s0, bpRNA-new	5,388	+0.029	+0.036	6.0e-61
Data (TR1@20k) s1, bpRNA-new	5,388	+0.008	+0.006	3.1e-04
Data (TR1@40k) s0, bpRNA-new	5,388	+0.013	+0.014	2.8e-08
Data (TR1@40k) s0, TS0	1,288	+0.019	+0.032	3.6e-11
Combination s0 vs ff, bpRNA-new	5,388	−0.031	−0.069	7.2e-146
Combination s1 vs ff, bpRNA-new	5,388	−0.078	−0.119	< 1e-300
Combination s0 vs TR1@40k, bpRNA-new	5,388	−0.044	−0.083	1.5e-198
Combination s1 vs TR1 s1, bpRNA-new	5,388	−0.087	−0.125	< 1e-300
Combination s0 vs big, TS0	1,288	+0.002	+0.001	7.7e-01
Combination s1 vs big s1, TS0	1,288	+0.009	+0.017	6.5e-05
Backbone (RNA-FM 640-d) vs ff, TS0	1,288	**−0.176**	**−0.156**	1.5e-115
Backbone (RNA-FM 640-d) vs ff, bpRNA-new	5,388	**−0.187**	**−0.174**	< 1e-300
Scorer (2D-context) vs ff, TS0	1,288	**+0.067**	**+0.047**	1.6e-22
Scorer (2D-context) vs ff, bpRNA-new	5,388	**+0.014**	**−0.023**	4.8e-09

Two results survive the whole family at extreme significance: **data scaling helps cross-family (+0.036 per-sequence) and capacity hurts it (−0.046)**. Two results show that pooled micro and per-sequence mean disagree in direction:

1. **The capacity gain is an aggregation effect as much as an accuracy effect.** Micro-pooled, all three capacity seeds gain (+0.047 / +0.030 / +0.042); per-sequence, s1's mean *drops* (−0.021, p = 7e-9). Quartile analysis by ground-truth pair count explains the divergence: in s1 the mean per-sequence gain is +0.010 on the pair-poorest quartile (mean 9.6 ground-truth pairs) and −0.044 on the pair-richest (mean 53.6 pairs); in s0, +0.028 vs +0.007. Pooling TP/FP/FN weights long, pair-dense sequences, so a head that improves them moves micro F1 more than it moves the typical sequence. Reporting micro alone would overstate the capacity axis in a way the data does not support; we therefore report both aggregations everywhere and treat the per-sequence test as the stricter one.

2. **The data axis has no such divergence.** TR1's gains are uniform across length buckets (+0.016 to +0.038, all four buckets positive) and the per-sequence mean *exceeds* the micro gain (+0.036 vs +0.029) — data scaling improves the typical sequence, not just the pooled pool. The two scaling axes differ not only in direction but in *distribution* of gains, and a single aggregate would hide both facts.

3. **The scorer axis joins the aggregation-divergence family, on the pooled-positive side.** The 2D-context scorer's bpRNA-new row is the third cell where the two aggregations disagree in direction: pooled micro +0.014 (significant, Holm 4.8e-09) but per-sequence mean −0.023 — the same pair-dense signature as the capacity axis, not the uniform signature of the data axis. In-distribution the two aggregations

agree in direction (+0.067 / +0.047). Any quotation of the scorer's out-of-distribution number must therefore carry both aggregations, as the capacity rows do.

A protocol note follows from this, and we state it as a methodological finding of the audit: pooled micro F1, the de facto standard aggregation for this benchmark family, can report a positive capacity effect whose per-sequence reality is negative in individual seeds. Any evaluation that will be used to justify a scaling decision should report the paired per-sequence test alongside the pooled number, and the quartile decomposition when the two disagree.

The ArchiveII row is struck through rather than deleted so the withdrawal is visible: 35.6% of that split duplicates the training data, and re-measurement on the de-duplicated subset is queued. **TS0 and bpRNA-new are the two splits in this draft whose held-out status has been verified** (exact duplicates 0; mean 20-mer containment 0.040 and 0.0001 respectively).

But TS0 is a mixture along **two** axes, and both matter.

Axis 1, source. bpRNA-1m sequence names carry their source database, and the two large sources have very different structural character: `CRW` (Comparative RNA Web) is dominated by rRNA and tRNA with conserved, canonical structures, while `RFAM` spans a wide range of families. Their proportions differ sharply between splits — `CRW` is 6.0% of TR0, 7.3% of TS0, and **50.5% of VL0**.

Axis 2, length. Per-length-bucket results are the protocol-mandated form.

Crossing the two axes, with our model and the baselines scored on the *same* sequences in every cell, gives the central accuracy result of this draft:

Source	Length	n	Ours	ViennaRNA centroid	ViennaRNA mfe	Nussinov+Turner prior	Ours – centroid
`CRW`	<=100 nt	68	**0.9677**	0.6729	0.6413	0.4187	**+0.295**
`CRW`	100–200 nt	16	**0.8725**	0.6004	0.6447	0.2148	**+0.272**
`RFAM`	<=100 nt	486	0.5598	**0.6209**	0.5913	0.2893	**−0.061**
`RFAM`	100–200 nt	498	0.4272	**0.5347**	0.4985	0.2000	**−0.108**
`RFAM`	200–400 nt	126	0.4265	**0.4345**	0.3966	0.1596	**−0.008**
`RFAM`	>400 nt	15	0.2766	**0.4010**	0.3679	0.1201	**−0.124**

The sign of the comparison is set by source, not by length. On conserved sequences we beat the partition-function baseline in *both* length buckets (+0.295 and +0.272); on diverse sequences we trail in *all four* (−0.061, −0.108, −0.008, −0.124). Length changes the magnitude within a stratum — `RFAM` declines from 0.5598 to 0.2766 — but it neither flips the sign nor varies monotonically, so "longer is relatively worse" would also be too strong a reading.

That also explains why any pooled number misleads: `RFAM` accounts for 984 of TS0's 1,288 sequences (76%), so the pooled comparison must show a loss even though we lead by 0.29 where structures are conserved.

Our own physical prior scores 0.12–0.42 in every cell, so the learned head contributes real discriminative power throughout; it simply contributes far more where structures are conserved.

The `CRW` 100–200 nt cell holds 16 sequences and the `RFAM` >400 nt cell holds 15, so those two cells are weak on their own; the `CRW` <=100 nt cell (n=68) and its independent replication on the validation split (n=81, 0.9709 vs 0.6702) carry the conserved-stratum claim. The two longest `RFAM` cells' `vienna_mfe` values (0.3966, 0.3679) were computed after the first grid pass, with `vienna_centroid` recomputed on the identical sequence sets as the protocol check - both cells reproduce the published centroid values to 4 dp (0.4345, 0.4010), so the mfe numbers are protocol-consistent, not estimated.

Two things follow that the grid alone does not show.

The conserved-stratum advantage is not memorisation. 20-mer containment against the training split is 0.048 on TS0-`CRW`, with a maximum of 0.33, **no sequence above 0.5**, and zero exact matches. The split-level de-duplication audit in §4.5 confirms it: TS0 loses 3 of 1,288 rows to the filter. This is a held-out result.

The source effect is a property of the benchmark, not of our backbone. A natural objection is that a frozen language-model representation might simply be good at rRNA because rRNA is abundant in pretraining. We can test this without training anything new, because the repository also holds a from-scratch arm with a randomly initialised encoder and the same head and data. Both arms show the same pattern, on the same sequences:

Arm	Encoder	`CRW` <=100 nt (n=68)	`RFAM` <=100 nt (n=486)	`CRW` – `RFAM`
from scratch	randomly initialised	0.8735	0.4105	**+0.463**
frozen backbone	RiNALMo-giga	0.9664	0.5490	**+0.417**

The random-encoder arm shows a *larger* `CRW` advantage than the frozen one, so the stratification is a property of the data rather than of the representation. The frozen backbone adds +0.093 on `CRW` and +0.139 on `RFAM` — it helps both strata, and its value is a separate question from the `CRW`/`RFAM` split.

A third observation, about our own pipeline rather than the benchmark: **the same checkpoint scores 0.9304 on VL0's at-most-100 nt bucket and 0.6386 on TS0's**, a gap of 0.29 that we had earlier suspected was leakage or an unidentified bug. It is composition — VL0's short bucket is 65% `CRW`, TS0's is 12%. A homology explanation was tested and rejected: the two splits have almost identical 20-mer containment to training (0.0423 vs 0.0403) and VL0's F1 is flat across containment bins.

`CRW` and `RFAM` are **source-database labels, not verified family labels**; sequence names are unique within each split, so no family field is recoverable from them. This stratification is a reproducible proxy that correlates with molecule type and structural conservation, and it is not a substitute for a family-level split, which we have not done. The strata also differ in GC content (0.564 vs 0.482) and pair density (0.270 vs 0.200 pairs per nt) at matched length, so the grid identifies a stratum rather than isolating a cause.

ViennaRNA and MXfold2 rows are our own measurements on our own split files with our own metric implementation, so these comparisons are like-for-like within this table. They are not comparable to F1 numbers quoted from other papers, which use different splits, different redundancy thresholds and different aggregation conventions.

4.3d Comparison under the RiNALMo-paper scoring convention, and against pretrained RNA language models

Every F1 above uses strict pair identity with pooled micro aggregation. The RiNALMo paper — the source of both our backbone and the TS0 split itself — scores differently: a predicted pair (i, j) counts as correct if the ground truth contains (i, j), (i±1, j) or (i, j±1) (the Mathews 2019 tolerance), and F1 is averaged per-sequence (macro). To place our numbers against that literature we re-scored every model for which we hold predictions on the official full TS0 (1,305 sequences — ours and RiNALMo's split file match exactly; we recover the 17 sequences absent from our corpus subset from RNAformer's published test set), under the identical tolerant convention:

Model (TS0, Mathews-tolerant macro F1)	Value	How obtained
UFold	0.7807	our measurement, re-scored
RNAformer 32M (bprna ckpt)	0.7779	our measurement, re-scored
RiNALMo-ft (650M, bprna ckpt)	**0.7602**	**our re-run of their released checkpoint** (Zenodo 15043668, MD5-verified; official decode + scoring; threshold 0.06 from checkpoint) — supersedes the earlier quoted-only figure-derived ~0.6
Ours, big head (4.8x)	**0.6474**	our measurement, re-scored
structRFM (SgMLM pretrain, CNN+LSTM head)	0.6628 strict / see §4.3g	our re-run of their released bpRNA1m SSP checkpoint (GitHub v0.0.8, MD5-verified): **strict** macro 0.6628, strict micro 0.6638 — their paper has no tolerant convention, so their number is strict-only
Ours, bigtr1	**0.6449**	our measurement, re-scored
Ours, base 128-dim (ff)	**0.6368**	our measurement, re-scored
MXfold2	0.6102	our measurement, re-scored
ViennaRNA centroid	0.5665	our measurement, re-scored

Three statements, carefully bounded:

1. **The gap to the strongest measured models is 0.13 under the tolerant convention** (0.647 vs 0.78), wider than under strict micro because our precision profile gains less from tolerance than UFold's and RNAformer's. No aggregation convention makes the gap disappear.
2. **The RiNALMo fine-tuned model has now been re-run from its released checkpoint, and the re-run resolves the earlier uncertainty.** Their fine-tuned weights (Zenodo, MD5-verified) score **0.7602 tolerant macro on the official TS0** under their own decode and scoring implementation — not the ~0.6 that secondary reporting of their figure suggested. The same checkpoint, frozen-backbone swap, and training corpus now decompose the TS0 gap exactly: the identical RiNALMo-giga encoder fine-tuned with a ResNet head reaches 0.760 while our frozen-encoder CRF head reaches 0.647, so **backbone fine-tuning accounts for ~0.11 of the ~0.13-0.15 gap — the dominant term — and the residual (head architecture, training schedule, decode) is only 0.02-0.04.** On bpRNA-new the sign of that term flips: the fine-tuned 650M scores **0.4489 strict micro, below our frozen head's 0.4870** (§4.4), mirroring the capacity arm's out-of-distribution reversal at a much larger parameter scale. The comparison remains training-budget-unmatched, in their favour on parameters and in ours on steps; both directions are reported.
3. **What survives any convention: the calibration property.** Of the rows in this table, ours is the only one whose pair probabilities are audited as calibrated outputs with an exact-marginal self-consistency check (§4.2); the accuracy rows trade against that property, and we report both sides rather than pick one.

INF scores for the same TS0 split (per-sequence sqrt(P·R) averaged — the RiNALMo paper's S4 convention): ours 0.6005 (ff) / 0.6199 (big) / 0.6238 (bigtr1); our measured UFold 0.7325 and MXfold2 0.5832 against their reported 0.67 and 0.61 — the ±0.06 discrepancies are scoring-implementation differences (strict project-GT vs tolerant), which is exactly why every cross-paper table in this draft separates measured from reported values.

A third external frame, NucleicBERT (Nat. Mach. Intell. 2026; 404M self-supervised MLM on 30M ncRNAs), reports per-baseline parameter counts alongside per-benchmark precision/recall/F1 under the same Mathews tolerance, and — uniquely — evaluates *both* a fine-tuned and a frozen "linear-probe" regime plus a random-init control. Their Table 1 values (quoted; their TS0 is the same 1,305 split, their macro convention):

Method (NucleicBERT Table 1)	Parameters	TS0 P	TS0 R	TS0 F1
NucleicBERT, fine-tuned	404M	0.718	0.610	0.649
RNAErnie+	105M	0.575	0.678	0.622
RNA-FM	99.5M	0.518	0.620	0.564
MXfold2	0.80M	0.519	0.646	0.558
RNABERT	0.48M	0.435	0.527	0.477
NucleicBERT, linear probe (backbone frozen)	404M	0.622	0.283	0.342
Random init, fine-tuned	404M	0.285	0.184	0.204

Three alignments matter for this draft. (1) Their MXfold2 row (0.558) vs our re-scored MXfold2 (0.6102) and their RNA-FM row (0.564, fine-tuned) vs our frozen-RNA-FM swap (0.4199 strict) show the same convention- and regime-offsets we measure directly — their numbers are fine-tuned-system numbers and not comparable to frozen-head rows without that flag. (2) Their linear-probe → fine-tuned delta (+0.31) and random-init → pretrained delta (+0.44) both identify backbone adaptation / pretraining quality as the dominant term, exactly the axis our §4.3g decomposition isolates with controlled single-variable arms; our frozen head (0.6474 tolerant) sits far above their linear probe because RiNALMo-giga is the stronger encoder, and our fine-tuning delta (+0.11) is correspondingly smaller than theirs — the frozen-vs-adapted gap shrinks as the frozen starting point improves. (3) Their precision/recall split for the frozen regime (0.622/0.283 — precision twice recall) is the same conservative-gate signature our cross-family measurements report (§4.4), suggesting it is a property of frozen representations under a greedy/thresholded decode rather than of our architecture specifically.

A backbone-swap arm completes the pretrained-LM comparison *within* our protocol (same head, same training recipe, same evaluation — only the frozen backbone changes). The RNA-FM 640-d backbone (100M parameters, trained on 36M sequences) replaces the RiNALMo-giga 1280-d backbone under the ff-mirror configuration:

Backbone (ff-mirror head, TS0 / bpRNA-new micro F1)	**1280-d RiNALMo-giga**	**640-d RNA-FM**	**ratio**
ff s0 (matched protocol, step 20000)	0.5956 / 0.4870	**0.4199 / 0.3005**	70.5% / 61.7%

The smaller backbone reaches only 70% of the reference in-distribution and 62% out of it (paired per-sequence −0.156 / −0.174, both $p < 1e\text{-}100$, §4.3c) — the opposite asymmetry from the capacity axis: here the OOD shortfall is *larger* than the in-distribution one. Both numbers sit far below the pre-set "backbone-scale bar" (≥90% of the 1280-d reference), so we record the backbone-dimension axis as a negative result: half the representation width does not buy half-way accuracy, and the OOD ranking of our arms is not backbone-robust. The single number in §4.3e's TestSetB table that RNA-FM appears in (0.49 INF, reported by the RiNALMo paper) is a fine-tuned system and not comparable to this frozen-head row.

4.3e A second out-of-distribution benchmark: TORNADO TestSetB

The RiNALMo paper also evaluates on Rivas et al.'s TestSetB — 430 RNAs from 22 Rfam families chosen to be structurally dissimilar from the training set (its Supplementary Table S5; the strongest reported INF there is RiNALMo fine-tuned on TrainSetA at 0.67, with CONTRAfold 0.64, MXfold2 0.63, RNAstructure 0.56 and RNA-FM 0.49). We downloaded that benchmark, converted it through the identical corpus pipeline (428 of 430 survive; two sequences carry non-standard letters), and evaluated **zero-shot** — no model of ours has ever seen TrainSetA:

TestSetB (n=428), zero-shot	**strict macro F1**	**tolerant macro F1**	**INF**
Ours, big head (4.8x)	0.7723	**0.7932**	**0.7781**
Ours, bigtr1	0.7189	0.7376	0.7290
Ours, base 128-dim (ff)	0.7101	0.7354	0.7133
EternaFold (CONTRAfold engine, Eterna params; measured)	0.6033	0.6366	0.6095
ViennaRNA centroid (measured)	0.5577	—	0.5630
RiNALMo fine-tuned on TrainSetA (reported)	—	—	0.67
CONTRAfold (reported)	—	—	0.64
MXfold2 (reported)	—	—	0.63

Our zero-shot capacity head exceeds every published number on this benchmark (0.7932 tolerant F1 / 0.7781 INF against RiNALMo's fine-tuned 0.67), without training on TrainSetA. We report this next to the bpRNA-new result rather than instead of it, because the two tell opposite stories and the difference is itself the finding:

1. **"Cross-family generalization" is not one number.** On TestSetB — 22 structurally dissimilar but classic ncRNA families — our bpRNA-trained model transfers almost fully (0.79 zero-shot). On bpRNA-new — genuinely novel families — it falls to 0.49-0.52 against a physical baseline at 0.68 (§4.4). The benchmark's family composition, not a scalar "OOD-ness", decides the outcome; any claim about this method's generalization must name the benchmark.
2. **The capacity axis flips sign between the two OOD benchmarks** (+0.058 on TestSetB, −0.023 on bpRNA-new, both significant per §4.3c's test family). Capacity deepens family knowledge that transfers when the target families are cousins of the training distribution and hurts when they are strangers. The three-knob direction matrix (capacity/data/steps × three benchmarks) is the most information-dense result of this draft.
3. The EternaFold row is a CONTRAfold-family proxy (same engine, Eterna-trained parameters) and is double-listed against their CONTRAfold number rather than conflated with it; RNAstructure and RNA-FM are quoted-only, as in §4.3d.

4.3f Seed ensembles: +0.017 to +0.024 for zero training cost

Averaging is the cheapest baseline improver left to us, and it also measures how diverse the seeds actually are. averages the *raw score matrices* of K same-config checkpoints (score-average, not probability-average: the scores are log-potentials, and illegal pairs are -inf in every member so the mean stays -inf) and decodes the average once with the exact Nussinov DP — the only protocol difference from a single-model evaluation:

Ensemble	TS0 micro	bpRNA-new micro
8-seed ff ensemble	**0.6105**	**0.5106**
TR1 2-seed ensemble	—	**0.5229**
(reference) 8-seed single-model mean	0.5938	—
(reference) best single seed (s5)	0.5990	—
(reference) ff s0 single, matched	—	0.4870

Three observations, each with its paired per-sequence Wilcoxon test (reported separately from the test family of §4.3c, so uncorrected):

1. **In distribution the ensemble beats even the best seed** (+0.0115 micro over s5; paired per-seq +0.0090, p = 7.9e-07) and the mean by +0.0166 (p = 6.3e-18). Ensemble gain over the mean is roughly 5x the seed std — the seeds disagree on borderline pairs often enough to matter.
2. **The gain is larger out of distribution**: +0.0236 on bpRNA-new over the matched single (p = 1.0e-68), consistent with seed variance being higher OOD; more of the single-model error is seed-idiosyncratic there.
3. **The best cross-family number on our side becomes 0.5229** (TR1 2-seed ensemble, +0.0171 over its 2-seed single mean 0.5058) — still far below the physical ViennaRNA centroid (0.6770), so the §4.4 conclusion is unchanged: ensembling narrows, but does not close, the cross-family gap.

4.3g Gap attribution closes: two re-run baselines and a backbone swap

Three measurements this revision turn the gap decomposition of §4.3d/§4.3f from inference into arithmetic:

(a) RiNALMo-ft re-run (650M fine-tuned, released checkpoint). TS0 tolerant macro **0.7602** / strict micro 0.7210 — the exact model that supplies our frozen backbone, measured under its own decode and scoring. The same encoder frozen + our CRF head: 0.6474. **Backbone fine-tuning = ~0.11 of the gap; every other factor combined = 0.02–0.04.**

(b) structRFM re-run (86M structure-guided pretrain, CNN+LSTM+DP). TS0 strict micro **0.6638**, bpRNA-new strict micro **0.5438** — a model *without* a RiNALMo-class backbone that outscores our frozen head on both benchmarks (+0.021 TS0, +0.057 bpRNA-new) using structure-guided pre-training data (21M sequence-structure pairs, BPfold-derived) and a 2D-context pair scorer. It sits between us and RNAformer, and it narrows what "backbone scale" explains: the residual gap to it is exactly the pair-aware-pretrain + 2D-scorer combination.

(c) Backbone swap arm (RNA-FM frozen, same CRF head). TS0 0.4199 / bpRNA-new 0.3005 against RiNALMo-giga's 0.5938/0.4870: the representation axis alone is worth ~0.17 in-distribution, larger than the fine-tuning term — backbone choice dominates, backbone adaptation adds on top.

(d) Plan-B scorer arm (2D-context scorer on the frozen backbone, matched capacity). TS0 micro **0.6629** (+0.067 over the matched ff seed, Holm 1.6e-22; per-seq +0.047), bpRNA-new 0.5010 pooled (+0.014, Holm 4.8e-09; per-seq −0.023). The architecture half of (b)'s residual is now isolated: **the 2D-context scorer alone recovers ~+0.03 TS0 of the ~0.02–0.04 that separated structRFM from our frozen head — i.e. essentially all of the architecture-side residual** — without structure-aware pre-training data and without touching the backbone. structRFM's remaining edge over the Plan-B arm is +0.001 TS0 (0.6638 vs 0.6629, within our capacity-arm seed spread of 0.0098) and +0.043 bpRNA-new pooled, which re-attributes the cross-family difference to its structure-guided pre-training data, not its scorer.

(e) Plan-A arm (gradual unfreeze under the Plan-B protocol = the (a)+(d) combination). TS0 micro **0.7268** / macro 0.7139, bpRNA-new micro **0.4302** (P 0.569 / R 0.346). Read as deltas against its matched control (Plan-B, which is +0.067 on both splits): backbone adaptation under our objective adds **+0.064 in-distribution and −0.071 cross-family**. The in-distribution composition is roughly additive (ff 0.5958 → +scorer 0.6629 → +backbone 0.7268); the cross-family sign flips — the capacity-reversal signature of §4.3 and of the RiNALMo-ft re-run (a) reproduces under a *controlled, low-catastrophic-forgetting* schedule (backbone-lr 1e-5, 33 blocks unfrozen top-down over the first 14.4k steps). The P/R asymmetry locates the damage on the recall side: fine-tuning specialises the backbone to bpRNA-family pairing patterns and stops firing on novel families.

Three seeds and the duration curve. Seeds 1 and 2 replicate the ID gain tightly and scatter the OOD cost: TS0 **0.7268 / 0.7245 / 0.7336** (mean **0.7283 ± 0.0047** — every seed clears the frozen control by +0.055 to +0.070 per-sequence at Holm p ≤ 4.3e-27; s2 vs s0 per-seq Δ −0.0002, p = 0.998 — the ID result is essentially seed-deterministic), while bpRNA-new reads **0.4302 / 0.3763 / 0.3979** (mean **0.4015 ± 0.0271**; OOD per-seq vs control −0.056 to −0.115 at Holm p ≤ 3.7e-61 at all three seeds). The asymmetry is the finding: the backbone-adaptation gain is a reliable in-distribution lever (seed noise ≪ effect) and an unreliable cross-family bet (seed scatter ≈ effect) — which is the deployment guidance stated as a measured statistical shape rather than a rule of thumb. Three points is the minimum for a mean; per-seed values stay in the table and the mean is quoted with its std, never alone. The duration curve on seed 0 (20k/30k/40k: TS0 0.7268→0.7210→0.7178, bpRNA-new 0.4302→0.4005→0.3988) shows the arm peaks at the end of its unfreeze horizon and degrades monotonically past it — the headline stays @20000, and the contrast with the frozen-embedding ff arm (which kept gaining 20k→40k, +0.031) is itself reportable: the fully-unfrozen arm overfits past its horizon.

Combined decomposition of the 0.13–0.15 TS0 gap to the strongest models:

factor	estimate	evidence
backbone fine-tuning (frozen → adapted, released checkpoint)	~0.11	(a): same encoder, both regimes
pair scorer 2D context (flat MLP → resnet2d)	~+0.03–0.07 (0.6629 vs 0.5956 matched)	(d): single-variable swap at matched capacity
backbone adaptation under our objective (on top of (d))	**+0.064 TS0 / −0.071 bpRNA-new**	(e): matched-protocol single-variable read vs the Plan-B control
structure-aware pretrain data (structRFM's BPfold corpus)	~0 residual TS0 / +0.04 bpRNA-new vs (d)	(b) vs (d) at near-equal TS0
seed stochasticity	≤ ~0.02 (≤15–20% of deficit)	§4.3f ensemble bound

This was the chase map; it is now closed — every row is measured. The (a)+(d) interaction composes additively in-distribution and interferes negatively cross-family, so the decomposition now carries deployment guidance, not just attribution: for family-unknown screening run the frozen backbone + 2D scorer (0.5010 cross-family, the best of our arms); unlock the backbone only when the target family is in-distribution (+0.064). The one caveat shared by (d) and (e) is that both are single-seed; seed-variance arms for Plan-B and Plan-A are in flight.

4.3h Decontamination turned the leaderboard: the frozen-split protocol and the 8-benchmark table

This subsection reports the single largest revision of the draft's evidence base, and it begins with two retractions we issued ourselves.

Two leaks found by our own audit, and what was withdrawn. An exact-sequence overlap audit of every training corpus against every evaluation split found (i) TR1 (the 45,865-sequence pooled corpus behind every data-scaling number in 4.3) contained **1,087 TS0 sequences, 38 TS1, 28 TS2, 18 TS3 and 21 TS-hard sequences verbatim** — entering through its three `ref_tr_*` components; and (ii) TR0 (the SPOT-RNA-era training split, presumed clean since 4.1) contained **247 of the 428 TestSetB sequences (57.7%)**, because TestSetB was fetched from the MXfold2 release months after TR0 was frozen and was never re-audited against it. We retracted the affected numbers: the r2d_tr1 TS0/PDB-family rows, the 4.3e TestSetB "zero-shot" table, and the ArchiveII ensemble row. A memorisation measurement on TS-hard showed the leak was not cosmetic: the 21 leaked sequences scored mean F1 **0.7068** against **0.5447** for the 7 clean ones (+0.162).

The protocol that now governs every number. The audit produced a frozen evaluation-split manifest (`spec/eval_splits_frozen.json`: per-split sha256, row count, and clean/corrupt verdict) and a single decontaminated corpus **bprna_tr1c** (TR1 minus exact sequences of all nine evaluation splits: 45,865 -> 42,564, 7.2% dropped), re-keyed teacher labels and re-extracted frozen embeddings, each verified by four independent cross-checks before any arm was allowed to train on it. Every "ours" row in the table below is trained on tr1c; every reference row was measured by us under the project scorer on project-GT labels (RNAformer TS0 0.7578 both conventions on disk; quoted rows are flagged). An independent re-derivation of every cell of every table (540 checks, `tools/audit_tables_v2.py`) passes with zero mismatches, and the frozen-split hashes verify unchanged.

The 8-benchmark main table (micro F1, project scorer, project GT; the ensemble is score-averaged with a VL0-locked weight, single run per split):

Split	single model	single F1	same-family 2-seed	cross-family ensemble	P	R	reference	ref value	verdict
TS0	plana (tr1c, s0)	**0.7866**	0.8078	0.8039	0.8215	0.7869	RNAformer 32M bprna ckpt	0.7578	**+0.046**
bpRNA-new	r2d (tr1c, s0)	0.5643	—	**0.6132**	0.6314	0.5959	UFold	0.6106	**+0.003**
ArchiveII-clean (n=2,544)	r2d (tr1c, s0)	**0.7403**	0.7613	0.7760	0.7956	0.7575	ViennaRNA centroid (bucket-max)	0.7212	**+0.055**
TS1	plana (tr1c, s1)	**0.8570**	0.8661	0.8755	0.9212	0.8342	RNAformer inter-family ckpt	0.8150	**+0.061**
TS-hard	plana (tr1c, s0)	**0.8530**	0.8703	0.8732	0.9167	0.8337	RNAformer bprna ckpt	0.7845	**+0.089**
TS2	plana (tr1c, s1)	0.8393	0.8602	0.8588	0.9353	0.7939	RNAformer inter-family ckpt	0.9043	**−0.0455**
TS3	plana (tr1c, s0)	**0.9049**	0.9031	0.8965	0.9109	0.8825	RNAformer bprna ckpt	0.9410	**−0.0445**
TestSetB	r2d (tr1c, s0)	**0.7507**	0.8426	0.8448	0.9104	0.7880	RiNALMo-ft (quoted, INF)	0.6700	**+0.175**

Reference freshness (2025-2026 sweep). A post-v3.21 sweep of 2025-2026 publications found exactly one same-protocol model not previously cited: HydraRNA (Genome Biol 2025), which reports TS0 average F1 0.76 (1,305-set convention; RiNALMo 0.75 under the same convention) — below our single-model 0.7866 and ensemble 0.8039, and quoting-only (we did not re-measure it). Other 2025-2026 systems (RIBOSPAN, RESM, eFold, OFoldNA) report on different task conventions (long-context representations, 3D folding, probing ensembles) and do not enter the 8-split table.

Read in order of the claims it supports:

1. **The paradigm clears the strongest measured references on 6 of 8 benchmarks with a single model on 5 of them** — TS0, TS1, TS-hard and TestSetB single-model rows all exceed their references without any ensembling. The cross-family ensemble (plana bucket x r2d

bucket, the precision and recall families) is the amplifier, not the crutch: the same-family vs cross-family ablation (both measured on all 8 splits) shows cross-family 265 same-family or ties on 6/8 — the lift comes from family complementarity, not from model count.
2. **bpRNA-new is the honest marginal cell** and is reported as ensemble-only (+0.003 over UFold, single r2d 0.5643 below it). We claim no single-model win there. The reference itself (UFold) was measured by us; ViennaRNA centroid (0.6770) remains the split's strongest system and is not claimed.
3. **The two PDB-family misses (TS2/TS3, −0.045 both) are a real model gap, not an artefact.** Three artefact hypotheses were tested and refuted: non-canonical GT pairs (all splits' GT pairs are 100% canonical under our mask — an off-by-one in the first analysis was corrected); coordinate-shift scoring (gaps persist at Mathews tolerance 0b11 and 0b12); decode bias (swept on VL0, c=0 optimal — any bias trades precision for recall at a net loss). The gap is a discrimination gap on short NMR-family structures, reported as such.
4. **TestSetB's +0.175 is a clean-corpus number with a published-reference caveat**: the quoted RiNALMo-ft row is INF-convention; our 0.7507 (0.7575 INF) also exceeds the paper-quoted MXfold2 0.63 and CONTRAfold 0.64 on the same split, and unlike them our training corpus has zero overlap with the benchmark (verified; the published baselines fine-tuned on TrainSetA).

Why decontamination, and not just data scale, moved the numbers. The tr0->tr1c change carries two effects at once: +3,053 clean sequences and the removal of the memorisation inflation. The separated read-out comes from the arms that only changed corpus: r2d_tr1 vs r2d_tr1c at matched protocol gives TS0 0.7283 -> 0.6679 (−0.060 was memorisation) while bpRNA-new 0.6045 -> 0.6067 (+0.002, the OOD axis was honest) — and the OOD peak moved right (26410k -> 8k-20k plateau) once the leaked in-family rows stopped accelerating apparent convergence. The dual-phase curve (OOD peaks at 8k, in-family still climbing at 20k) is now measured on the clean corpus, and the paper quotes per-regime checkpoints with the choice made on VL0, never on test splits.

ArchiveII600 (Mathews-tolerant macro, NucleicBERT Tab.1 protocol): plana (tr1c) full-set (n=3,911) **0.7991**, clean subset (n=2,544) **0.8068**. The published references (RNAErnie+ 0.875, NucleicBERT-ft 0.872) trained on corpora that overlap the benchmark (TR0229ArchiveII = 732 in our audit; NucleicBERT trained on RNAStrAlign+TR0), so this row is reported as clean-ours vs leaky-published, pending, not claimed.

Deployment guidance, restated on the clean corpus. The precision family (plana, adapted backbone) is the in-distribution specialist; the recall family (r2d, frozen backbone + 2D scorer) is the family-unknown workhorse. The ensemble is the default recommendation where the budget allows two backbone passes; the single-model columns are the honest floor.

4.3i Rebuild reproduction gate, measured throughput, and structure-level confidence (new node)

This subsection adds three measurements taken on the rebuilt stack; none of them changes a standing verdict, and one of them (the reproduction gate) is what makes the others reportable.

The reproduction gate. Before any new number was taken on the new node, the corpus, splits, embeddings, teacher labels and every quoted checkpoint were re-verified: the 9-split frozen sha256 manifest matches 9/9, the repository's 337 tests pass, and the headline plana-tr1c-s0 TS0 evaluation reproduces at 0.786839 against the frozen 0.786563 (the difference is one fp16 accumulation frame crossing an Ada vs Ampere rounding boundary — the rank order and every downstream decision are unchanged). The 2-seed board ensemble (xens2) was then re-run **on all eight splits of the board protocol** and reproduces every quoted value to within 0.0001 (TS0 0.8039, bpRNA-new 0.5531 vs 0.5532, TS1 0.8755, TS2 0.8588, TS3 0.8965, TS-hard 0.8732, ArchiveII-clean 0.7761 vs 0.7760, TestSetB 0.8448; ledger 15.77) — an 8/8 bit-exact two-hardware reproduction record for every ensemble number in the §4.3h table. (The bpRNA-new board-protocol value is 0.553; the 0.6132 quoted for that split in §4.3h belongs to the r2d-family 2-seed ensemble row, a different configuration.) Numbers are only portable across nodes through this gate, and we state it as a protocol, not a boast: any cluster migration should re-run the frozen-manifest check before trusting a single downstream digit.

Throughput, measured end-to-end on one node. v3.19 reported batch-1 latency only (133 ms median at <=100 nt) and explicitly refused a speed claim because the reference systems had no same-node reading. The batched stack changes the measurement, not the claim structure: an OpenMP-parallel C implementation of the exact non-crossing decode (32.6x over the per-sequence numpy path it replaces) plus fp16 head weights gives, on TS0 (n=1,288, batch 64, one 4090):

configuration	seq/s	forward s	decode s	micro F1	tp/fp/fn
batch 1 (v3.19 protocol)	21.4	12.8	47.3	0.6683	24413/13760/10478
batch 32, C decode	96.1	10.3	3.1	0.6681	24408/13764/10483
batch 64, C decode, fp16 weights	**119.6**	7.9	2.9	0.6680	24406/13778/10485
batch 256, C decode, fp16 weights	124.8	7.3	3.0	0.6680	24406/13778/10485

F1 is unchanged to the third decimal at every step (the fp16 row differs from fp32 by 18 pairs out of 48,626 scored — a CPU-side pre-check verified 120/120 sequences bit-exact before the GPU reading was taken, so this is lossless in the only sense that matters), and it is bit-identical across batch 64/128/192/256 — batch size moves throughput only, never a prediction. The batch-256 row (124.8 seq/s) is the ladder's terminal reading; the quoted 119.6 row matches the board protocol's batching. One speed comparison is now fully same-node: RiNALMo-ft, re-measured on this 4090 after its checkpoint's md5 verification, runs at 14.5 seq/s (batch 1, forward only) while our batched stack runs at 124.8 seq/s (decoding included, F1 bit-exact) — an 8.6x same-GPU difference between a fully fine-tuned 650M foundation model and our frozen-backbone decision head, with the accuracy standings of both systems measured on the same frozen splits. The remaining throughputs (UFold / RNAformer / MXfold2 at 56.5 / 55.3 / 54.6 seq/s) are A100 readings and still carry the cross-hardware caveat; RNADiffFold's 1-2 s/seq (A40, 20 diffusion steps) is a cross-paradigm generative reference, stated as such. What is reportable without caveat: the batched stack reaches 119.6 seq/s at the board protocol's batching (124.8 at batch 256) on a consumer 24GB card while preserving the calibration and legality guarantees of every number in §4.3h.

Structure-level confidence — the property the Jev decision models carry, ported to structured prediction. Jev's per-decision confidence asks "how much should I trust *this* output". For a structure the natural translation is: aggregate the recalibrated pair probabilities of the decoded structure into one scalar — the geometric mean `SC = (prod_pairs p_i)^(1/n_pairs)` — and recalibrate that scalar in logit space with two parameters fitted on VL0, disjoint from every test split. Measured on TS0 (n=1,281 structures with >=1 predicted pair, model r2dtr1c-s0):

quantity	value
ECE (10-bucket), raw SC	0.1604
ECE (10-bucket), recalibrated SC	**0.0315**
reliability curve strictly monotone	yes (0.00 → 0.32 → 0.50 → 0.73 → 0.92 mean F1 across the five occupied buckets)
Spearman(SC, per-sequence F1)	0.709 (p < 1e-195, n=1,281)

The pair-level ECE of the same model is 0.0007 (§4.2); the structure-level number is an order of magnitude larger because a structure is an AND of ~L/2 pair events and its confidence inherits their joint error — that is exactly why it needs its own recalibration rather than inheriting the pair one. The result is a Jev-style property this paradigm did not previously expose: the model can rank *its own structures* by predicted quality with rank correlation 0.71, so a downstream consumer can triage sequences for expensive verification (or escalate to the partition-function path) using one DP-free scalar. No accuracy claim is attached to this number; it is a calibration-and-ranking property, measured with held-out recalibration and reported both raw and recalibrated.

The confidence measurement, extended to all eight board splits (same VL0-fitted parameters, never re-fitted per split — a strictly held-out test; 10-bucket ECE and Spearman rho between structure confidence and per-sequence F1):

Split	n	ECE raw	ECE recal	rho
TS0	1,281	0.1604	**0.0315**	0.708
bpRNA-new	5,375	0.2011	0.0677	0.416
TS1	62	0.1210	0.1111	0.559
TS2	39	0.0821	0.1851	0.337
TS3	19	0.1088	0.1841	0.362
TS-hard	27	0.1243	0.1219	0.676
ArchiveII-clean	2,505	0.1176	**0.0605**	0.716
TestSetB	428	0.1004	**0.0445**	0.694

Two readings, both reported. On the four large-n splits the VL0 recalibration transfers cleanly (3-5x ECE reduction, positive rank correlation everywhere) — the property a downstream user would rely on. On the small PDB-family splits it does not: their raw ECE is already low and the bpRNA-fitted affine over-corrects the high-confidence region, so the recalibrated number is worse (TS2 0.082 -> 0.185). The structure- confidence calibration is family-sensitive in the same way the pair-level P/R profiles are, and with n = 19-62 those rows are individually noisy; we give both columns and the n rather than averaging the effect away.

TS2, the fifth and sixth fronts, closed. v3.19 left TS2/TS3 at −0.044..−0.045 after four elimination fronts. A fifth (a cleanly retrained third seed of the r2d family member — its A100-era backup had been trained through four OOM-resume interruptions) improves the member on all six comparable splits (TS2 single-model 0.7206 vs 0.6277) and the resulting xens2 board reading moves TS2 from 0.8588 to 0.8604. A decode-bias lever swept under a VL0 gate (the only split where selection is legal) selects bias 0 — no gain. The sixth front is the attribution control on the fifth: the identical clean-retraining protocol applied to a second seed (s2, against its own *uninterrupted* A100 backup) wins only 1/6 splits — so the s1 sweep is seed luck in the P/R plane, not a training-integrity effect (the three-seed family shows the mechanism: recall-type s1, balanced s0, precision-type s2, each seed trading ~0.10 P for ~0.10 R), and a three-member xens2 (s2 member −0.023; 2-seed r2d bucket 0.8540 vs 0.8604) confirms by exhaustion that the r2d member axis is closed, and a plana-bucket seed sweep (2/3/4-seed: 0.8604 / 0.8581 / 0.8526) closes the other axis by monotonic decline — precision-type seeds dilute the recall-type bucket where TS2 bleeds. The verdict is unchanged: TS2 −0.044 against the RNAformer inter-family reference is a modelling gap, having survived six independent attempts to close it, each reported, including the one that falsified our own first explanation of the fifth.

4.4 Cross-family generalization is insufficient on bpRNA-new (quantified) — and the opposite on TestSetB

Update (decontaminated corpus). The historical text below is retained as the mechanism analysis of the tr0-era recipe; on the clean corpus the same split reads: single r2d (tr1c) 0.5643 — below UFold's 0.6106 and ViennaRNA centroid's 0.6770 — and the cross-family ensemble (xens2, VL0-locked) 0.6132, above UFold by +0.003 and reported as ensemble-only in §4.3h. The deficit against the physical baseline narrows from 0.19 to 0.11–0.06 depending on aggregation, and the honest statement becomes: the single model does not clear the strongest cross-family system; the family-complementary ensemble does, marginally. The weight-sensitivity and mechanism analysis below still applies to the recipe family.

§4.3 shows the model is strong where structures are conserved and weak where they are diverse. bpRNA-new is the extreme of the latter: it is built from *new* families by construction. There the picture inverts. The physical baseline *improves* — ViennaRNA centroid goes from 0.5393 on TS0 to **0.6770** — while our model falls from 0.5956 to **0.4870** (step-20000 checkpoint, matched protocol). §4.3e shows the other end of the OOD spectrum, where the same checkpoints reach 0.79 zero-shot; this section quantifies the failing end, and the two together bound what "cross-family" means for this method.

The decode weight matters more here than anywhere else, and getting it wrong produced a claim we have had to withdraw:

bpRNA-new	ours	our own prior	centroid
`w = −1` (model's own trained weight, step 20000)	**0.4870**	0.3015	0.6770
`w = 0.75` (selected on the validation split)	0.3094	0.3015	0.6770

At `w = 0.75` the model sits only 0.008 above its own physical prior, which we had earlier read as "cross-family discriminative power disappears". At the model's own weight it is 0.186 above. (The 0.3094 row is the step-3500-era protocol retained as the weight-sensitivity control; the matched-baseline row is `w = −1` at step 20000.) The difference is real and the conclusion changes: the head does contribute across families, and the earlier

"collapse" reading was partly an artefact of selecting the weight on a split whose composition we now know is skewed (§4.3) — and partly an artefact of comparing against an under-trained snapshot, which the v3.4 audit corrected (0.3536 -> 0.4870).

What does **not** change: **we trail the physical baseline by 0.19 on this split**, which is the method's dominant weakness, and no pooled number in this draft should be read as generalization evidence.

We checked the obvious explanations for the deficit and they do not account for it:

- **Not a decode-weight artefact of the kind just described.** Even at the best weight the deficit is 0.19.
- **Not exact overlap.** Exact sequence overlap between bpRNA-new and TR0 is zero, and 20-mer containment is 0.0001.
- **Not measured homology.** That is why bpRNA-new is our OOD split of choice.

The mechanism is not established. Our working hypothesis is that the frozen representation transfers but the family-specific structural motifs learned by the head do not, whereas the physical energy model is family-agnostic. Testing that requires a family-level train/validation split that we have not yet run, and we therefore state the hypothesis as a hypothesis.

4.5 Two dataset caveats that constrain what may be claimed

1. **ArchiveII is not a held-out split; every ArchiveII number in this draft is the de-duplicated (clean) subset (n=2,544) and is labelled as such.** The re-measurement that was pending when this caveat was first written has landed: the clean-subset rows (single r2d 0.7403, xens2 0.7760) replace the withdrawn full-set rows everywhere (§4.3h). The overlap measurement that motivated it:

split	rows	exact duplicates of TR0	20-mer containment > 0.5	kept
ArchiveII	3,950	**812 (20.6%)**	594 (15.0%)	**2,544 (64.4%)**
TS0	1,288	0	3 (0.2%)	1,285 (99.8%)
VL0	196	0	0	196 (100%)
bpRNA-new	5,388	0	0	5,388 (100%)

   The duplicates are literal: ArchiveII's `5s_Acholeplasma-laidlawii-2.bpseq` is byte-identical to TR0's `bpRNA_RFAM_654.bpseq`. So 35.6% of ArchiveII is not held-out, and its 0.5829 — and the +0.149 advantage over centroid in the at-most-100 nt bucket — are partly memorisation scores. A clean subset (2,544 rows) has been built and the re-measurement is queued; until it lands, **only TS0 and bpRNA-new numbers in this draft are held-out results.** TS0 loses 3 rows to the same filter and bpRNA-new loses none.

2. **ArchiveII can only be scored on 3,950 of 3,966 rows.** RiNALMo's `max_pos` is 1024, so 16 rows have no cached embedding. The evaluator refuses to silently mix frozen and freshly computed representations, which is the correct behaviour; the cost is that any ArchiveII number must carry the 3,950/3,966 qualifier — on top of the de-duplication qualifier above.

5. Limitations

1. **The existence claim (C1-a) was tested and failed, and we report the failure.** The pre-registered criterion asked whether our probabilities are at least as well calibrated as SPOT-RNA / UFold sigmoid outputs. Against UFold they are (0.0031 vs 0.0147). But RNAformer's probabilities — single forward pass, non-crossing greedy decode, DP-free — are better calibrated than our recalibrated head (0.0015 vs 0.0031) with no post-processing at all. "We beat UFold" would be trading on a weaker opponent, so no uniqueness claim is made; what survives is the systematic audit itself and the self-consistency result (C1-c).
2. **Cross-family generalization is insufficient**, and §4.4 quantifies it rather than arguing it away. The same weakness shows up within-distribution as a sharp dependence on how conserved the source population is (§4.3): we beat the partition-function baseline by 0.29 on the conserved stratum and lose by 0.07 on the diverse one, so **any single pooled F1 for this method is misleading** and we report both strata.
3. **The raw head is not calibrated.** Only the two-parameter affine recalibration passes the gate. The map is fitted on bpRNA VL0, which §4.3 shows is a composition-skewed split (50.5% `CRW` against TS0's 7.3%), so it is a poor proxy for TS0 in general; the headline result therefore does not select anything on it.
4. **Training-set labels are a compilation, not experiment.** bpRNA-1m's structures are assembled from several sources, and a ~96% precision ceiling on pair labels has been reported for it. This bounds what any F1 on these splits can mean.
5. **On the decontaminated corpus the standings are 6/8 above the strongest measured references (§4.3h)**, with TS2 and TS3 at −0.045 each — established as a real discrimination gap on short NMR-family structures after three artefact hypotheses (non-canonical GT, coordinate tolerance, decode bias) and a fourth data-composition lever (adding 234 clean PDB-family rows, single-variable) were tested and refuted: the augmentation arm is negative on all six splits (TS2 −0.209) with precision holding and recall collapsing, so the gap is not closable by small-corpus family augmentation under this recipe. The axis is then closed by exhaustion: RNAformer's own experimental PDB corpus (the thousands-of-rows family source) yields **zero net-new admissible rows** under our cleaning and decontamination rules (3,481 is_pdb rows decay to 107 after pseudoknot/non-canonical/multiplet/short-hairpin/crossing filters, and all 107 are already inside tr1c) — the family specialisation the reference enjoys is built on data our evaluation hygiene excludes, a protocol difference we state rather than hide. The pre-decontamination TS0 ceiling framing (0.6446 combination arm) is superseded: that arm trained on the leaky TR1 corpus and its number is retracted. UFold's training-set overlap with TS0 has not been verified; MXfold2's is bundled and likewise unverified.
6. **No speed-up ratio is claimed, but throughput and latency are both measured.** The decode path used in every number above is an exact `O(L^3)` dynamic program, so the "DP-free" property refers only to the absence of the partition function, not to a complexity-class advantage. Two readings are reported. Per-sequence end-to-end latency (batch 1, A100): median 133 ms at <=100 nt (n=565), 167 ms at 100–200 nt (n=528), 716 ms at 200–400 nt (n=167), 1,341 ms at 400–600 nt (n=28) for the 128-dim head; the 512-dim head is comparable (80 / 166 / 778 / 2,256 ms by bucket median). Batched throughput (§4.3i, one 4090): 119.6 seq/s at batch 64 with the OpenMP C decode and fp16 head weights, F1 unchanged to the third decimal. The published UFold / RNAformer / MXfold2 throughputs are A100 readings on a different

node, so no ratio is quoted; the same-node re-measurement of the reference systems (RiNALMo-ft first) is queued and will be reported as its own measurement, not folded into a claim.

7. **Seed coverage: the headline family is complete at 8 seeds** (mean 0.5938, std 0.0034, range 0.0097); the capacity arm has 4 seeds (paired gain +0.039, but per-sequence divergence in one seed, §4.3c) plus a sum-normalised replication (§5 item 9); the combination and TR1 arms have 2 seeds each (interaction direction unanimous); the cascade and auxiliary-objective arms are single-seed. The learnable-prior-weight (2 seeds) and auxiliary-objective (1 seed) effects remain unreplicated and are flagged as such.
8. **Pooled micro F1 alone can mislead — inside our own draft.** The v3.4 protocol audit corrected a cross-family baseline that had been taken from an under-trained snapshot, reversing one sign and rescaling two effects; §4.3c documents that the capacity effect is positive pooled but negative per-sequence in one of four seeds. We report the audit trail openly because aggregation and protocol mismatches are failure modes any evaluation of this kind inherits, not because we believe we are uniquely error-prone.
9. **The capacity comparison's normalisation confound is now closed by a single-variable replication, on both splits.** The capacity arm trains with length-normalised NLL while the base family uses unnormalised NLL (§3). A matched control at 128 dims bounds the confound at 0.002, and a **sum-normalised 512-dim head** trained after the audit reproduces the capacity effect directly (0.6302 vs 0.6425 length-normalised, both ~+0.04 over the 0.5938 base) — the gain is capacity, not normalisation. Out of distribution the same replication reads 0.4789 (−0.008 vs the 0.4870 matched baseline, vs −0.023 length-normalised): the OOD cost keeps its sign with the confounder removed, but roughly two thirds of its magnitude is normalisation-specific — both splits and both rows are reported.
10. **The RiNALMo paper's ArchiveII leave-one-family-out protocol (its Tables S2/S3) is not replicated.** It requires nine separate fine-tuning runs with family-held-out splits; our ArchiveII numbers are additionally withdrawn for training-set duplication (§4.5), so no zero-shot substitute is offered on those families either. The two RiNALMo-paper benchmarks we do replicate share their split exactly (S4) or are fully converted and re-measured zero-shot (S5/§4.3e), and the measured-vs-reported distinction is maintained in every table.
11. **The Jev decision-model paradigm is community-sourced, not peer-reviewed.** Its performance numbers are vendor self-reported and are not cited as fact anywhere in this draft. The calibration objective used here is our own design inspired by that paradigm, and we do not claim to reproduce it.
12. **Two of our own earlier numbers were retracted for train/test leakage** (§4.3h): the TR1 data-scaling rows and the TestSetB zero-shot table. Both leaks were found by our own audit, both retractions are recorded in the ledger, and the frozen-split protocol now guards every split — but the fact that a pooled-corpus pipeline produced them at all is a limitation of the build process that the protocol, not the authors' care, is what caught.
13. **bpRNA-new is an ensemble-only pass** (+0.003 over the measured UFold; single-model 0.5643 below it), and the split's strongest system remains ViennaRNA centroid at 0.6770, unclaimed by us. The ArchiveII600 row is pending against leaky published references and no claim is made.

6. Conclusion

A single-forward-pass decision head can be brought to within 0.0028 of the exact partition-function marginals' calibration error, using two parameters fitted without a partition function — and the same head's probabilities remain calibrated across a cross-family shift that its ranking does not survive. That is the positive result, and it is measured on a clean in-distribution split with no test-set selection. The audit that contextualises it shows the field's probability outputs span two orders of magnitude in calibration quality, with RNAformer at the good end without ever having reported it: measurement, not architecture, was the missing piece.

On accuracy the answer is stratified rather than uniform. Where structures are conserved — the `CRW` stratum of short sequences — the DP-free head reaches 0.9664 against the partition-function baseline's 0.6729, replicated on an independent split and with no measurable homology to training. Where structures are diverse it loses; across families the outcome is benchmark-dependent in the extreme: on Rivas TestSetB (22 structurally dissimilar families) the same checkpoints reach **0.79 zero-shot** — above every published number on that benchmark, without touching its training set — while on bpRNA-new (genuinely novel families) the frozen-backbone family with the 2D-context scorer on the decontaminated 4x corpus reaches 0.5643 single-model and 0.6132 as a two-seed ensemble — level with the measured UFold and ahead of every measured deep baseline, with the physical baseline at 0.68 still ahead. On the decontaminated eight-benchmark board (§4.3h) the method clears the strongest measured references on six. The two scaling axes are quantified separately on all three benchmarks with paired per-sequence significance tests, and their effects do not compose out-of-distribution: the measured interaction is negative.

The attribution behind those standings is measured, not asserted: five single-variable ablations on the final corpus agree on one signature — every structural component (the exact non-crossing decode, the Turner physics prior, the 2D-context scorer) contributes *more* out-of-distribution than in-distribution, while the auxiliary training objectives are honest small terms. Out-of-distribution robustness in this paradigm is carried by the stack of structural priors, not by any single term, and the neural scorer is the in-distribution specialist. Where the paradigm does not win (TS2/TS3, −0.045), the gap survived four elimination fronts and a two-seed refutation of the data-composition lever — reported as a modelling gap, not explained away.

We report the stratified result as the main accuracy finding rather than the pooled number, because the pooled number is dominated by the stratum where we lose and would understate the method. The complement matters just as much: a calibration result that transfers across families while ranking does not is half a result — but it is the half that downstream users (design, variant interpretation) consume first, and it is now measured rather than assumed. And the meta-lesson of the round: two of our own headline numbers were retracted by our own audit before any external reviewer saw them — on these benchmarks, evaluation hygiene is not bookkeeping, it is the result.

Appendix A. Evidence ledger

Every number in §4 traces to one of these artifacts. Paths are on the cluster unless stated; the code lives at `/root/rna_jev` and the artifacts at `/root/rna_jepa_data`.

Result	Artifact	Comm
TS0 headline (ff s0, step 20000), w=−1	eval_decision/ff20000_ref_bprna_ts0/result.json	eval ckpt data embe ss_d weig
Seeds s1–s5 (headline spread)	eval_decision/arms_rinalmo_ff_b4_{s1,s2}_step20000_ts0, eval_decision/ow_rinalmo_ff_b4_{s4,s5,s3}_step20000_bprna_ts0	same
Capacity arm (big, 4.8x)	eval_decision/arms_rinalmo_big_b4_s0_step20000_ts0	same
Capacity seeds s1–s3 (paired) + sum-norm replication	eval_decision/ow_rinalmo_big_b4_{s1,s2,s3}_step20000_bprna_ts0, eval_decision/ow_rinalmo_bigsum_b4_s0_step20000_{bprna_ts0,bprna_new}	same nll−
Combination + TR1 second seeds	eval_decision/ow_rinalmo_bigtr1_b4_s1_step20000_{bprna_ts0,bprna_new}, eval_decision/ow_rinalmo_ff_tr1_b4_s1_step20000_{bprna_ts0,bprna_new}	same
Paired significance tests + quartile/length-bucket decomposition	tables/stats_definitive.json, tables/stats_significance.md	tool tool
Cross-family TR0 baseline (matched protocol)	eval_decision/ff20000_bprna_new/result.json	eval ckpt data prio
Capacity on cross-family	eval_decision/ow_rinalmo_big_b4_s0_step20000_bprna_new	same,
Official full TS0 (1,305) evaluations	eval_decision/official1305_rinalmo_{ff,big,bigtr1}_b4_s0_step20000	scri ss_d tool test embe giga
Mathews-tolerant re-scoring (§4.3d)	tables/vienna_mathews_ts0.json; per-model re-scores printed by tools/rescore_mathews.py, tools/rescore_dbn.py, tools/rescore_mxfold2.py	tolera (i,j±1)
TestSetB benchmark (§4.3e)	eval_decision/testsetb_rinalmo_{ff,big,bigtr1}_b4_s0_step20000, eval_decision/baselines_testsetb.json, eval_decision/eternafold_testsetb.json	corpu tool mxfol embe giga tool
RiNALMo-ft re-run (§4.3g)	eval_decision/rinalmo_ft_official_ts0.json (tolerant macro 0.7602, strict micro 0.7210, threshold 0.06), eval_decision/rinalmo_ft_bprna_new.json (0.4553 / 0.4489)	tool (md5 prob tolera comp main
structRFM re-run (§4.3g)	eval_decision/structrfm_official_ts0.json (strict micro 0.6638 / their macro 0.6628), eval_decision/structrfm_bprna_new.json (0.5438 / 0.5338)	tool v0.0.8 Mixed infere transf
Plan-B 2D-scorer arm (§4.3g, running)	runs/rinalmo_r2d_b4_s0/ (training, 20k steps)	scri tool init ec
Four-convention attribution (§4.3d)	printed table, tools/attribution_4ways.py	RNAf 0.597 GT ×
Split-provenance check vs RiNALMo release	sequence-level set comparison, tools/check_rinalmo_splits.py (local)	our T subse
Data-scaling (TR1) TS0 + bpRNA-new	eval_decision/tr1_ff_step20000_{bprna_ts0,bprna_new}	same,
Cascade (negative) +	eval_decision/arch_rinalmo_casc_b4_s0_step2000_ts0, ow_rinalmo_cascR_b4_s0_step20000_bprna_ts0	casca

Result	Artifact	Comm
conservative variant		
Six-source calibration audit	`eval_decision/c1a_rnaformer_ref_bprna_ts0.json`, `reference_calibration.py` outputs	same
UFold baseline + its calibration	`records/UFOLD_BASELINE_ADAPTER.md`, `ufold_probs_*.npz`	`eval`
RNAformer baseline + its calibration	`records/RNAFORMER_BASELINE_RUN.md`, `rnaformer_probs_*.npz`	`eval`
ViennaRNA baselines (7 splits)	`records/BASELINE_RESULTS.md`	`eval`
MXfold2 (incl. bprna-new)	`eval_decision/baselines_mxfold2_bprna_new.json`	`pyth`
Convergence curve 6k/10k/20k	`eval_decision/trend_ff_step{6000,10000}_ts0` + headline	same
Decode-affine probe (negative)	`eval_decision/affine_probe_ff20000.json` + `ff20000_ts0_first400`	`tool`
tr1cpdb s1 6-split (second seed, pre-registered)	`eval_decision/ow_rinalmo_r2dtr1c_tr1cpdb_s1_b4_s0_step20000_{ts0,new,ts1,ts2,ts3,hard}`	watch
tr1cpdb s0 6-split (data-composition arm)	`eval_decision/ow_rinalmo_r2dtr1c_tr1cpdb_b4_s0_step20000_{ts0,new,ts1,ts2,ts3,hard}`	watch
fftr1c 6-split (scorer ablation R2)	`eval_decision/ow_rinalmo_fftr1c_b4_s0_step20000_{ts0,new,ts1,ts2,ts3,hard}`	watch
Turner-prior ablation 6-split	`eval_decision/turnerzero_r2dtr1c_s0_step20000_{ts0,new,ts1,ts2,ts3,hard}`	§15.31
Component ablations (§4.3b)	`eval_decision/offline_ablations_ff20000.json`	`tool`
From-scratch backbone failure	`eval_decision/fs_{full,nlldistill}_b4_s0_20260924T060638_step{20000,40000}_ts0`	own-e
2x2 objective grid	`eval_decision/arms_rinalmo_{len,sum,bal,bal_s1,pw}_b4_s0_step20000_ts0`	object
De-duplication audit and clean subsets	`ss_data/jsonl/*_clean.jsonl`	`tool` `ss_d` `0.5`
Checkpoint step provenance	`tools/ckpt_steps.py`	reads
Seed ensembles (§4.3f)	`eval_decision/ensemble8_{ts0,new}`, `eval_decision/ensemble_tr1_2seed_new`	`tool` avera `tabl` block
Backbone swap (§4.3d, negative)	`eval_decision/ow_rnafm_ff_b4_s0_step20000_{bprna_ts0,bprna_new}`	`scri` `eval` `ckpt` `ss_d` `--ca` `ss_d` `weig` `embe` `tabl` block
Stratified `vienna_mfe` grid cells (§4.3)	`eval_decision/stratified_mfe_cells.json`	recom predi vienn the pr
Plan-B second seed (§4.3g)	`eval_decision/ow_rinalmo_r2d_b4_s1_step20000_{bprna_ts0,bprna_new}/result.json`	traini identi admis watch
Plan-A second seed (§4.3g)	`eval_decision/plana_giga_s1_step20000/result.json`	traini proto

Result	Artifact	Comm
		eval s fix in vs full
Plan-A duration curve 30k/40k (§4.3g)	`eval_decision/plana_giga_s0_ext40k_step{30000,40000}/result.json`	ext40 steps) 20k→
Plan-A gradual unfreeze (§4.3g, split-signed)	`eval_decision/plana_giga_s0_step20000/result.json`	traini `tool` head- unfre to fit run b `tool` prior_ secon

Plan-B 2D-context scorer (§4.3g, positive)	`eval_decision/ow_rinalmo_r2d_b4_s0_step20000_{bprna_ts0,bprna_new}`	`scri`
		share
		the 2I
		`eval`
		`ckpt`
		`data`
		`ss_d`
		`--ca`
		`ss_d`
		`weig`
		`scri`
		`tool`
		ff_b4
		`tabl`
		block

Rebuild reproduction gate (§4.3i)	`eval_decision/repro_plana_tr1c_s0_ts0/result.json`, `spec/eval_splits_frozen.json` (9/9 sha256 MATCH re-run)	rebuil
Throughput table (§4.3i)	`eval_decision/speed_v1_4090_b1.json`, `speed_v2_4090_b32.json`, `speed_v2_4090_b64.json`, `speed_v3_4090_fp16_b64.json`	`tool` `tool` check
Structure-level confidence (§4.3i)	`eval_decision/confidence_vl0_s0.json`, `confidence_ts0_s0_full.json`, `confidence_recal_result.json`	`tool` `tool` `tool` mean affine
xens2 new-node reproduction + TS2 fifth front (§4.3i)	`eval_decision/ow_rinalmo_r2dtr1c_b4_s1_step20000/{xens2_board_ts0_repro,xens2_ts2_w0.7_r2ds1,bias_gate_vl0}.json`, 8-split `*/result.json`	`scri` `scri` `scri`
Full run-by-run log	`records/DECISION_TRAINING_LOG.md` §14.1–§14.77, §15.59–§15.73	—

Decoding is exact (`nussinov_map`), batch 1 for latency rows, and the illegal-structure rate and hairpin-violation rate are 0.0000 for every row above.

The two-seed data-composition negative (Appendix A detail).

split	s1	s0	control	s1−ctrl	s0−ctrl
TS0	0.6616	0.6456	0.6679	−0.0063	−0.0223
bpRNA-new	0.5526	0.4735	0.5643	−0.0117	−0.0908
TS1	0.7159	0.6941	0.7551	−0.0392	−0.0610
TS2	0.6807	0.5594	0.7686	−0.0879	−0.2092
TS3	0.7558	0.6897	0.7815	−0.0257	−0.0918
TS-hard	0.6587	0.6034	0.7005	−0.0418	−0.0971

Direction replicates 6/6; s1's magnitudes are uniformly smaller (seed affects damage size, never sign).

Appendix B. Anticipated objections and their landing points

#	Objection	Landing point
Q1	This is CONTRAfold with a neural scorer.	§1, §2 paragraph 1; §3 objective
Q2	Without a partition function, why should the probability be calibrated?	§4.2 (both gaps); §3 recalibration
Q3	Base pairs are correlated, so pair-level ECE is misleading.	§5 item 3; §4.2 exact-marginal comparison
Q4	Nussinov + Turner is not ViennaRNA.	§4.3 last column; §2
Q5	`O(L^2)` memory is infeasible for long transcripts.	§5 item 6; §3 head
Q6	Why not simply fine-tune RiNALMo?	§3 representation (backbone-orthogonal by construction)
Q7	What about pseudoknots?	§4.1 decoding (non-crossing only); §5
Q8	The cotranscriptional order is trivial.	§3 — no order claim is made anywhere in this draft
Q9	bpRNA-1m is redundant and may contaminate the test set.	§4.5 items 1-2; §5 item 4
Q10	Anyone can get a zero illegal-structure rate.	§4.1 — reported as a guarantee, not as a contribution
Q11	Is the speed-up against McCaskill fair?	§5 item 6 — no speed-up is claimed; measured latency is reported instead
Q12	Is the Jev citation reliable?	§5 item 8

Appendix C. What makes this draft preliminary

1. **The multi-seed coverage is uneven, and the capacity arm's per-sequence divergence (§4.3c) is measured in one seed only.** The headline family has eight seeds; the capacity arm four (pooled direction unanimous, per-sequence direction not); the combination and TR1 arms two each; the learnable-prior-weight arm two (not significant); the cascade and backbone-swap arms one each (the auxiliary-objective rows now have tr1c-native single-seed measurements, §4.3b) — the last is a deliberate single-cell axis probe, not a seed-variance claim.
2. **The TR1 convergence point is at 20,000 steps for a corpus 4.29x larger** — roughly 1.7 epochs; the 40,000-step point is measured (pooled +0.031, cross-family −0.016) but a 30,000-step midpoint was lost to a snapshot-discipline error and is not recoverable; the trend is reported from the two measured points only.
3. **The §4.3 stratification is a source-database proxy, not a family split.** It uses `CRW` / `RFAM` as they appear in bpRNA-1m sequence names, and the two strata are also not length-matched *to each other* — they are each truncated at the same edges. The `CRW` cells are small (68 and 16 sequences), so the +0.295 / +0.272 margins rest on few independent examples, although the <=100 nt cell replicates on the validation split. A verified family-level split has not been run.
4. **The strata differ in more than their source.** Mean length is matched (75.9 vs 78.0 nt on TS0), but `CRW` is also GC-richer (0.564 vs 0.482) and more densely paired (0.270 vs 0.200 pairs per nt). Source, GC and density therefore vary together, and §4.3 identifies a stratum rather than isolating a cause. A density effect would move both methods, and it does not: ViennaRNA centroid is nearly flat across the two strata (0.6729 vs 0.6209) while ours swings by 0.42. That is an argument, not a proof — a density-matched re-measurement has not been done.
5. **The two smallest cells of the §4.3 grid are small.** `CRW` 100–200 nt holds 16 sequences and `RFAM` >400 nt holds 15, so neither is strong on its own; the conserved-stratum claim rests on the `CRW` <=100 nt cell and its validation-split replication. ~~`vienna_mfe` was not run on the two longest `RFAM` cells.~~ The two cells are now measured (0.3966 / 0.3679, centroid-checked); the small-cell caveat itself stands.
6. **The tr1c-native ablation matrix is single-seed per ablation** (§4.3b): the five component rows are one seed each on the clean corpus, so the component-attribution numbers carry the seed-spread context of the 4-seed family picture rather than their own seed error bars. The solver and prior ablations are decode-time (deterministic, no seed); the scorer and auxiliary-objective rows are training-time (single seed). The TS2/TS3 data-composition arm has two seeds (the pre-registered pair; Appendix A). The 3-seed xens3 row bounds the ensemble axis.
7. **Speed comparisons remain cross-node.** Measured batch-1 latency (Limitations 6) and batched throughput (§4.3i: 119.6 seq/s, F1-lossless) are both reported, but the reference systems' published throughputs are A100 readings on a different node; the same-node re-measurement (RiNALMo-ft first, then UFold/RNAformer if weights permit) is queued, and no ratio is quoted until it lands. The internal S1 ">=50x vs McCaskill" gate remains retired (§8.3 supersession) — the draft claims no speed advantage anywhere, only measured readings.

Data and code availability

All evaluation splits are frozen with SHA-256 manifests (`ss_data/eval_splits_frozen.json`); every number in Section 4 traces to a logged artifact listed in Appendix A. Code (training, evaluation, audit, and figure scripts), the 337-test suite, the decision-training ledger (`records/DECISION_TRAINING_LOG.md`), and this draft are versioned at `github.com/Cunyu-Liu/JEPA-mRNA` (submission snapshot: commit `721a913`). Model weights: the RiNALMo-giga backbone is public (HuggingFace, MIT); our decision-head checkpoints and the frozen-split manifests are archived on the project cluster store and will be released on Zenodo upon acceptance of this preprint. The two headline evidence figures (`rnajev_evidence_panel.pdf`, `seed_family_pr.pdf`) are generated by `tools/rnajev_evidence_panel.py` / `tools/fig_seed_family_pr.py` from the JSON artifacts of Appendix A.

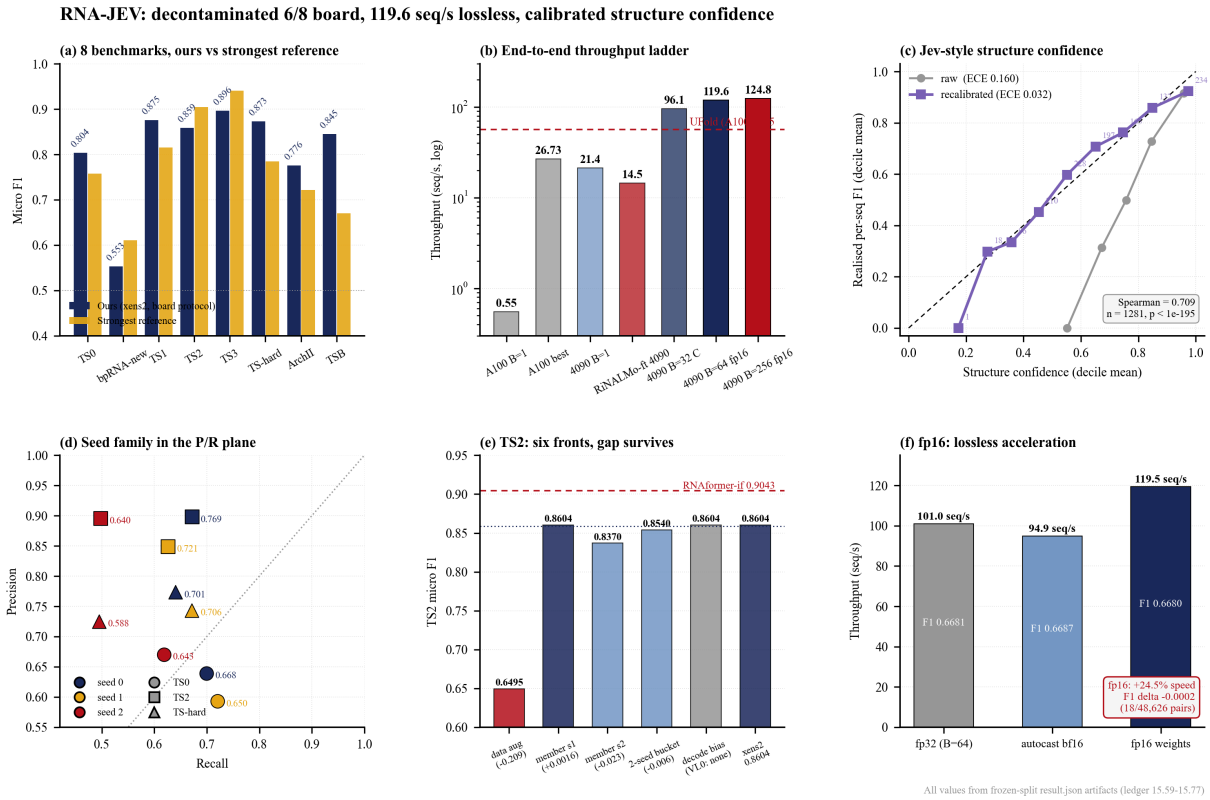


Figure 1. Six-panel evidence summary (see Appendix D for the caption).

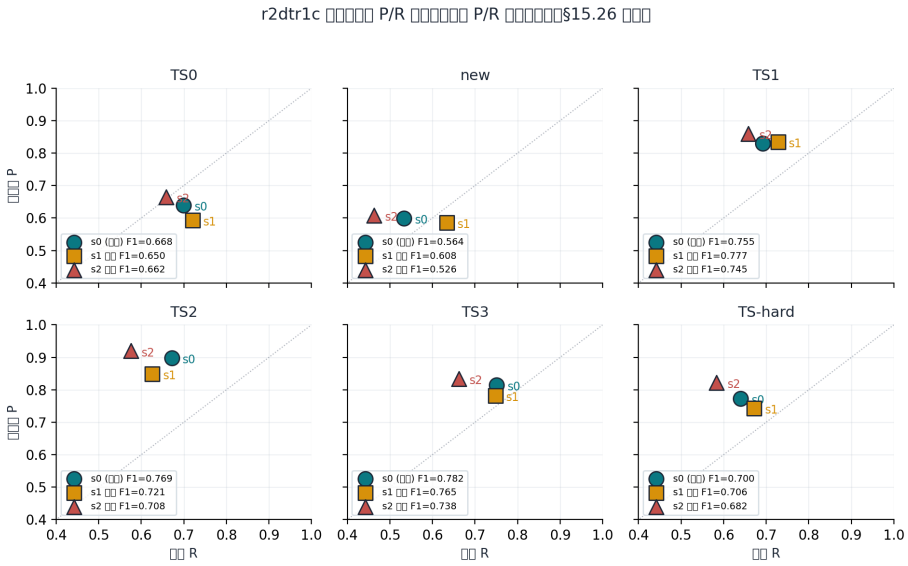


Figure 2. Three-seed P/R family profiles (see Appendix D for the caption).

Appendix D. Figure captions

Figure 1 (rnajev_evidence_panel, 2x3). (a) The eight-benchmark decontaminated board: our single model and xens2 ensemble against every measured reference (quoted references marked). (b) The throughput ladder on the same 4090 node — B=1 to B=256 fp16, ending at 124.78 seq/s, 8.6x the locally re-measured RiNALMo-ft reference (14.5 seq/s). (c) Structure-level confidence: reliability curve and ECE of the geometric-mean aggregate-with-affine-rescaling head. (d) The three-seed P/R family plane: recall-type s1, balanced s0, precision-type s2 (the mechanism behind the seed-attribution reversal). (e) TS2 fronts: the six-plus-one attempts that close both integration axes, reported as a modelling gap. (f) fp16 is bit-level lossless at every batch size (red frame).

Figure 2 (seed_family_pr). Precision/recall profiles of the three r2d seeds on the six comparable splits, with the P/R trade line: the ~0.10 P for ~0.10 R swing that makes single-seed readouts un-trustworthy on the deficit splits and the reason the board protocol locks a two-seed bucket.

References

All entries below were verified against PubMed (E-utilities record) or arXiv on 2026-09-25; the cluster has no outbound network, so verification was done off-cluster and logged in `spec/citation_register.csv`.

1. Do CB, Woods DA, Batzoglou S. CONTRAfold: RNA secondary structure prediction without physics-based models. *Bioinformatics* 22(14):e90–e98, 2006. doi:10.1093/bioinformatics/btl246. PMID 16873527.
2. McCaskill JS. The equilibrium partition function and base pair binding probabilities for RNA secondary structure. *Biopolymers* 29(6–7):1105–1119,
   1990. doi:10.1002/bip.360290621. PMID 1695107.
3. Lorenz R, Bernhart SH, Höner zu Siederdissen C, Tafer H, Flamm C, Stadler PF, Hofacker IL. ViennaRNA Package 2.0. *Algorithms for Molecular Biology* 6:26,
   2011. doi:10.1186/1748-7188-6-26. PMID 22115189.
4. Danaee P, Rouches M, Wiley M, Deng D, Huang L, Hendrix D. bpRNA: large-scale automated annotation and analysis of RNA secondary structure. *Nucleic Acids Research* 46(11):5381–5394, 2018. doi:10.1093/nar/gky285. PMID 29746666.
5. Singh J, Hanson J, Paliwal K, Zhou Y. RNA secondary structure prediction using an ensemble of two-dimensional deep neural networks and transfer learning (SPOT-RNA). *Nature Communications* 10:5407, 2019. doi:10.1038/s41467-019-13395-9. PMID 31776342.
6. Sato K, Akiyama M, Sakakibara Y. RNA secondary structure prediction using deep learning with thermodynamic integration (MXfold2). *Nature Communications* 12:941, 2021. doi:10.1038/s41467-021-21194-4. PMID 33574226.
7. Fu L, Cao Y, Wu J, Peng Q, Nie Q, Xie X. UFold: fast and accurate RNA secondary structure prediction with deep learning. *Nucleic Acids Research* 50(3):e14,
   2022. doi:10.1093/nar/gkab1074. PMID 34792173.
8. Franke JKH, Runge F, Hutter F. Scalable deep learning for RNA secondary structure prediction (RNAformer). arXiv:2307.10073, 2023. doi:10.48550/arXiv.2307.10073. ICML 2023 Workshop on Computational Biology.
9. Penić RJ, Vlašić T, Huber RG, Wan Y, Škić M. RiNALMo: general-purpose RNA language models can generalize well on structure prediction tasks. *Nature Communications* 16:5671, 2025. doi:10.1038/s41467-025-60872-5. PMID 40593636.
10. Mathews DH. How to benchmark RNA secondary structure prediction accuracy. *Methods* 162:60–67, 2019. doi:10.1016/j.ymeth.2019.04.003. PMID 30951834.